Pioneering Transformers: Pretraining + Fine-Tuning

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University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - Attention transformed: birth of a new architecture
  - Key idea: self-attention
  - Building a transformer
  - Pioneering model for machine translation
- Assignments (Canvas):
  - Problem set 3 (last one!) due Friday
  - Lab assignment 3 (last one!) to be released on Thursday (no class that day)
- Questions?

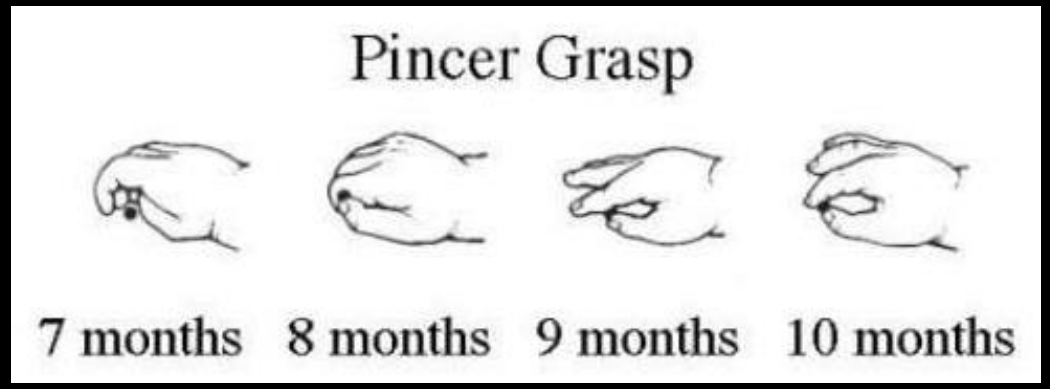
Today's Topics

- Revisiting learning: from solving tasks to building representations that generalize
- The recipe: large-scale pretraining + lightweight fine-tuning
- Multimodal test: can the recipe replace CNN-RNN solutions for vision-language tasks?
- Speech test: can the recipe replace engineered solutions for speech recognition?
- Vision test: can the recipe replace CNNs for image classification?
- Programming tutorial

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Learning Evolution: Basic Skills to Tasks



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/



e.g., from birth, your hands learn general coordination, building a foundation for later adaption to many tasks

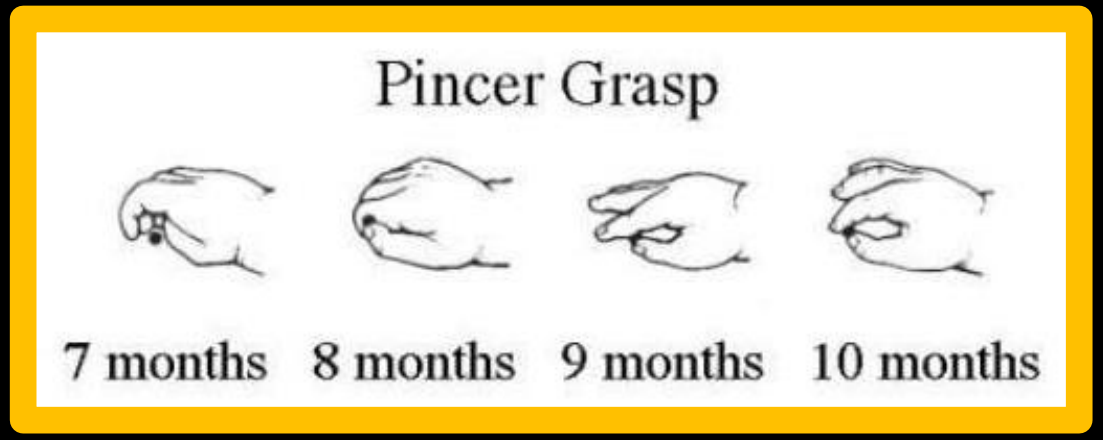
Deep Learning Status Quo: Task-Specific Focus



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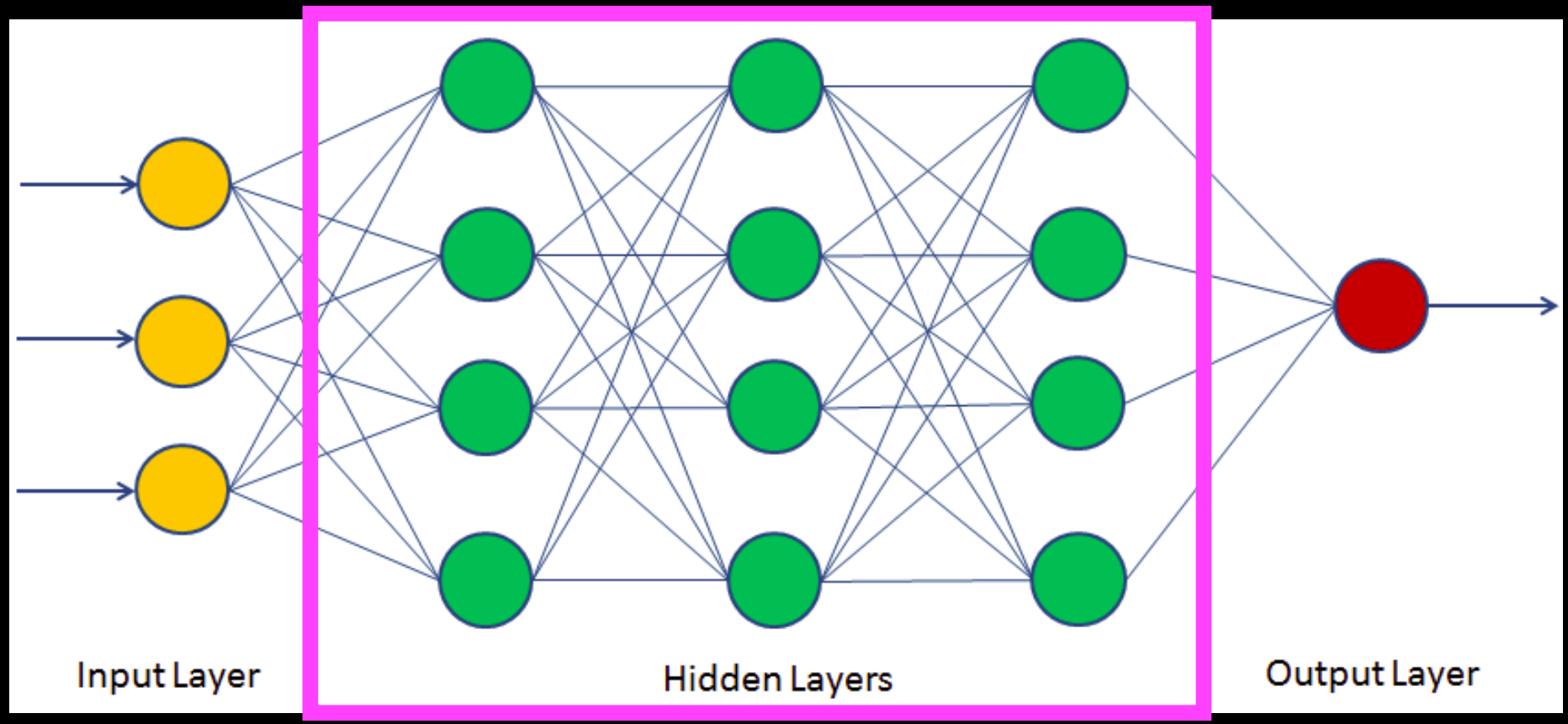
Idea: Focus on Building Representations that Can Be Reused for Many Tasks



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Idea: Focus on Building Representations that Can Be Reused for Many Tasks



https://www.datacamp.com/community/tutorials/neural-network-models-r

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Pioneering Methods: GPT and BERT

(Technical Report, 2018)

Improving Language Understanding by Generative Pre-Training

Alec Radford OpenAI
Karthik Narasimhan OpenAI
Tim Salimans OpenAI
Ilya Sutskever OpenAI

(NAACL, 2019)

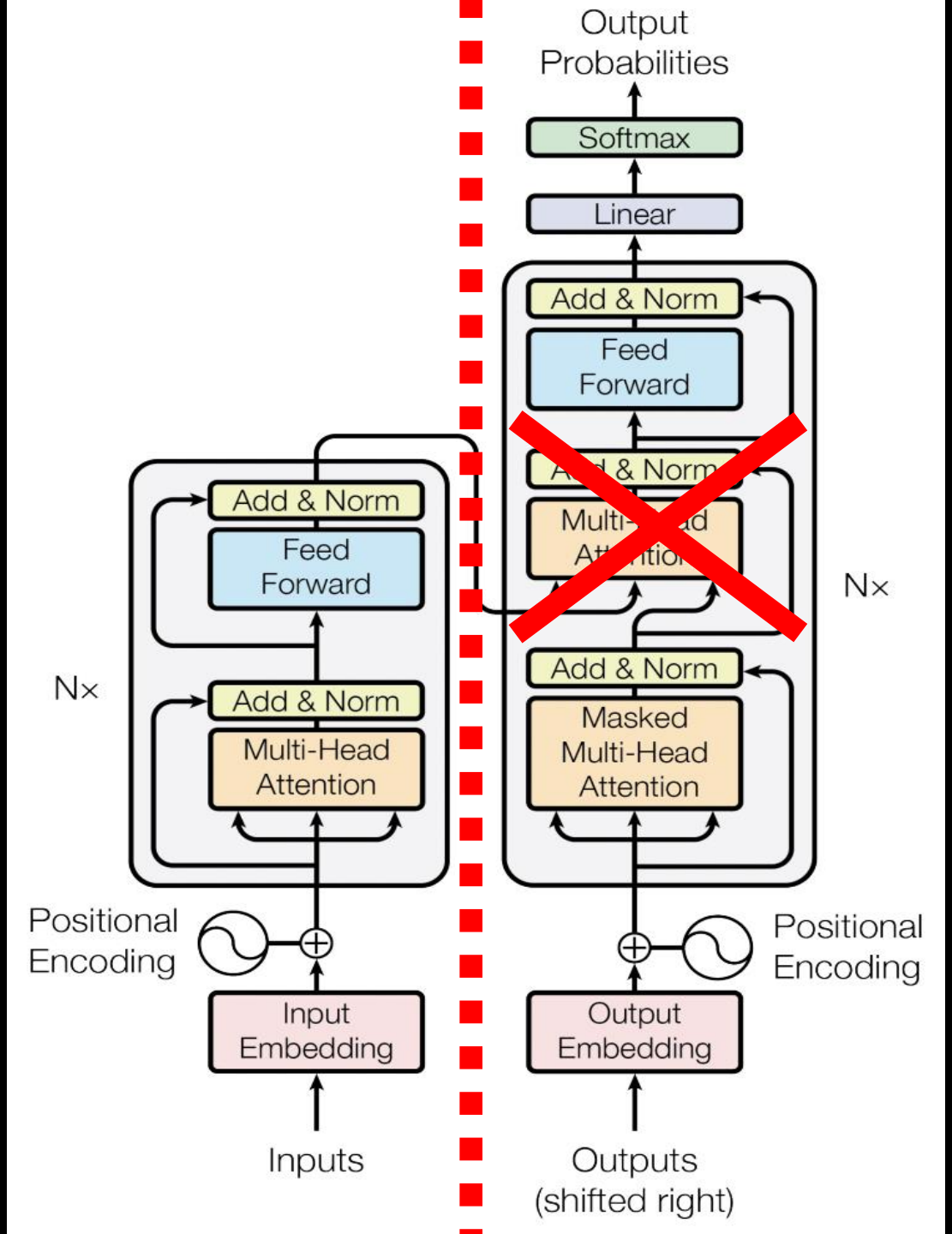
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin **Ming-Wei Chang** **Kenton Lee** **Kristina Toutanova**
Google AI Language

What architecture is this?

BERT

(encoder)



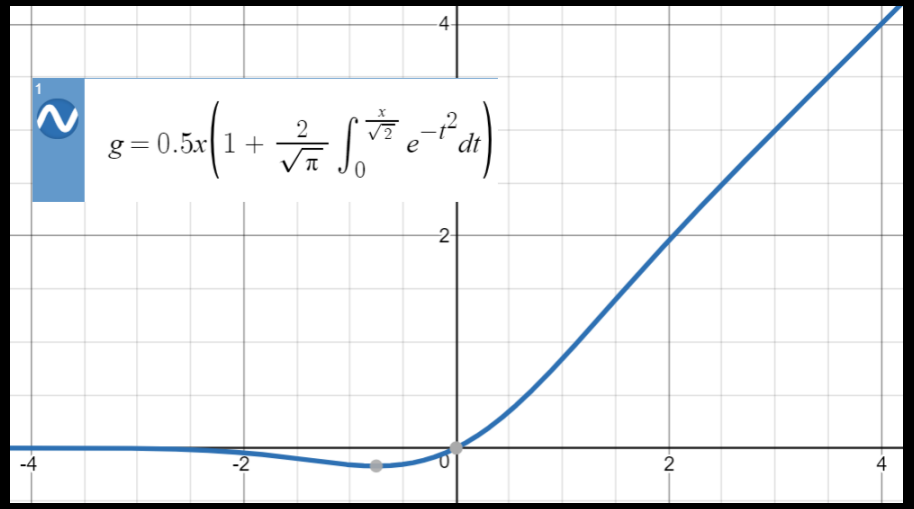
GPT

(decoder)

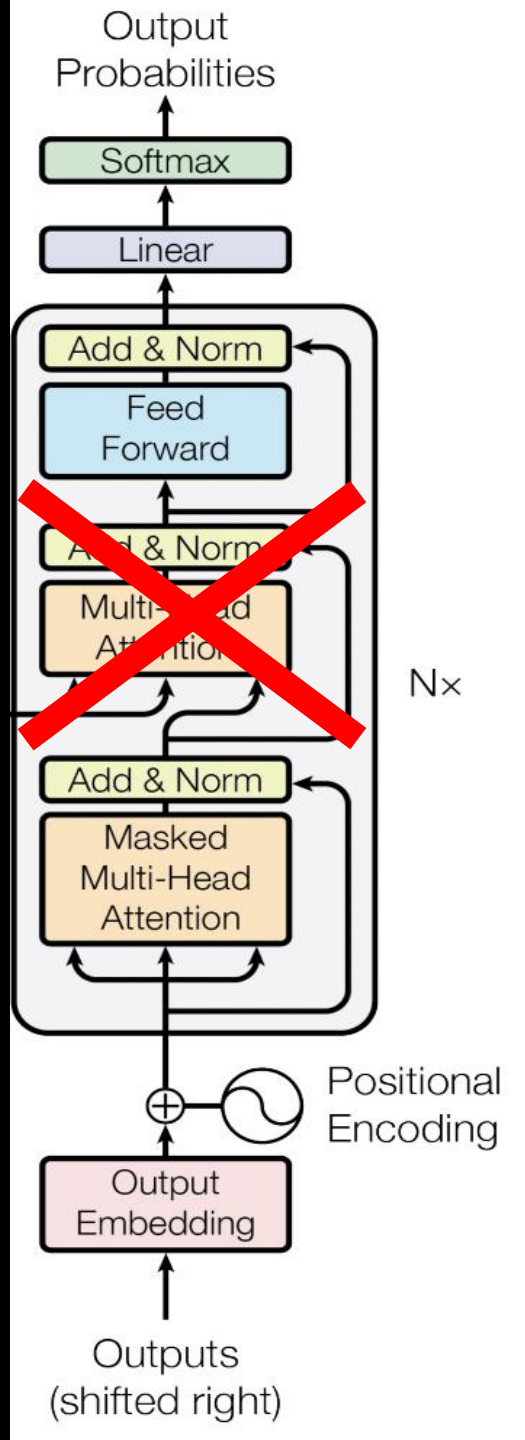
Vaswani et al. Neurips 2017.

Implementation details:

- Input encoding: byte-pair for 40,000 token vocabulary
- Position embeddings learned
- 12 layers (Nx)
- 12 heads in masked attention
- GELU activation function in feed forward layer



https://datascience.stackexchange.com/questions/49522/what-is-gelu-activation



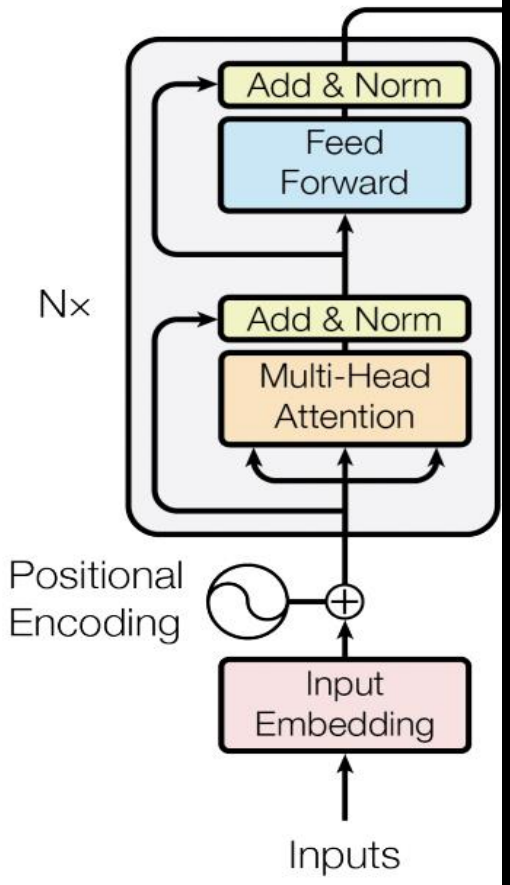
GPT

(decoder)

Vaswani et al. Neurips 2017.

BERT

(encoder)

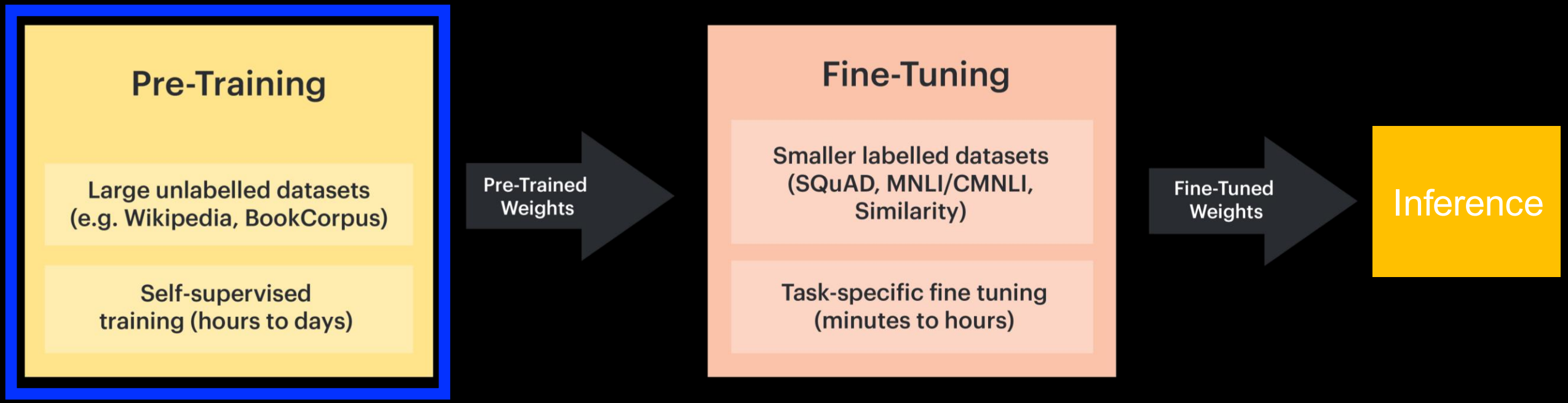


Implementation details:

- Input encoding: WordPiece for 30,000 token vocabulary
- Position embeddings learned
- 12 and 24 layers tested (Nx)
- 12 and 16 attention heads tested
- GELU activation function in feed forward layer

Vaswani et al. Neurips 2017.

Approach

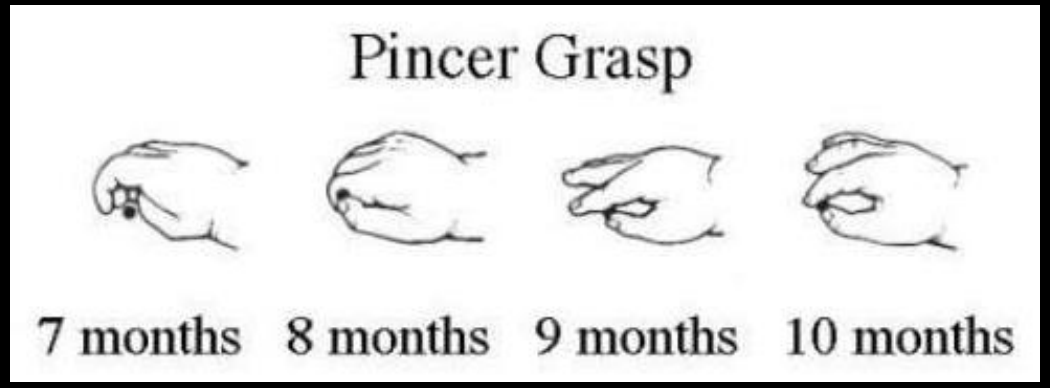


Since VERY challenging to collect large-scale human-annotated datasets, we don't!

https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

Approach

Self-supervised pretraining was not a new idea
(e.g., predict self with autoencoder to learn visual features; learn word embeddings)



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/



The new insight: a large transformer, trained on massive data, can achieve state-of-the-art results on many downstream tasks with minimal fine-tuning!

Key Question: What Learning Objective Yields the Best Representation? E.g.,

GPT = Generative Pre-Training

- predict next word given previous ones (unidirectional, left to right)

e.g., I arrived at the bank after crossing the _____

BERT = Bidirectional Encoder Representation from Transformer

- predict masked token AND predict if one sentence follows a second sentence

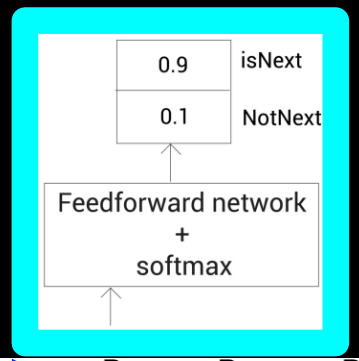
e.g., I arrived at the bank [MASK] crossing the river. My pants were wet.

BERT Implementation Details

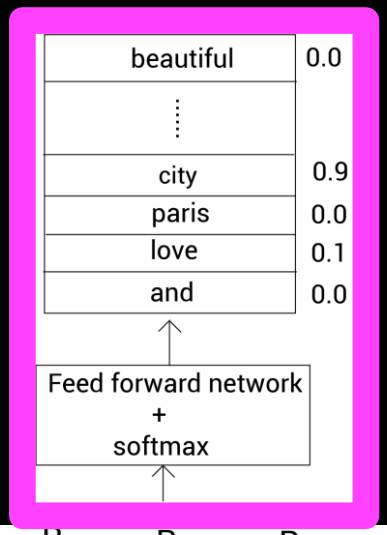
2 tasks:

- predict if one sentence follows a second sentence
- predict masked token

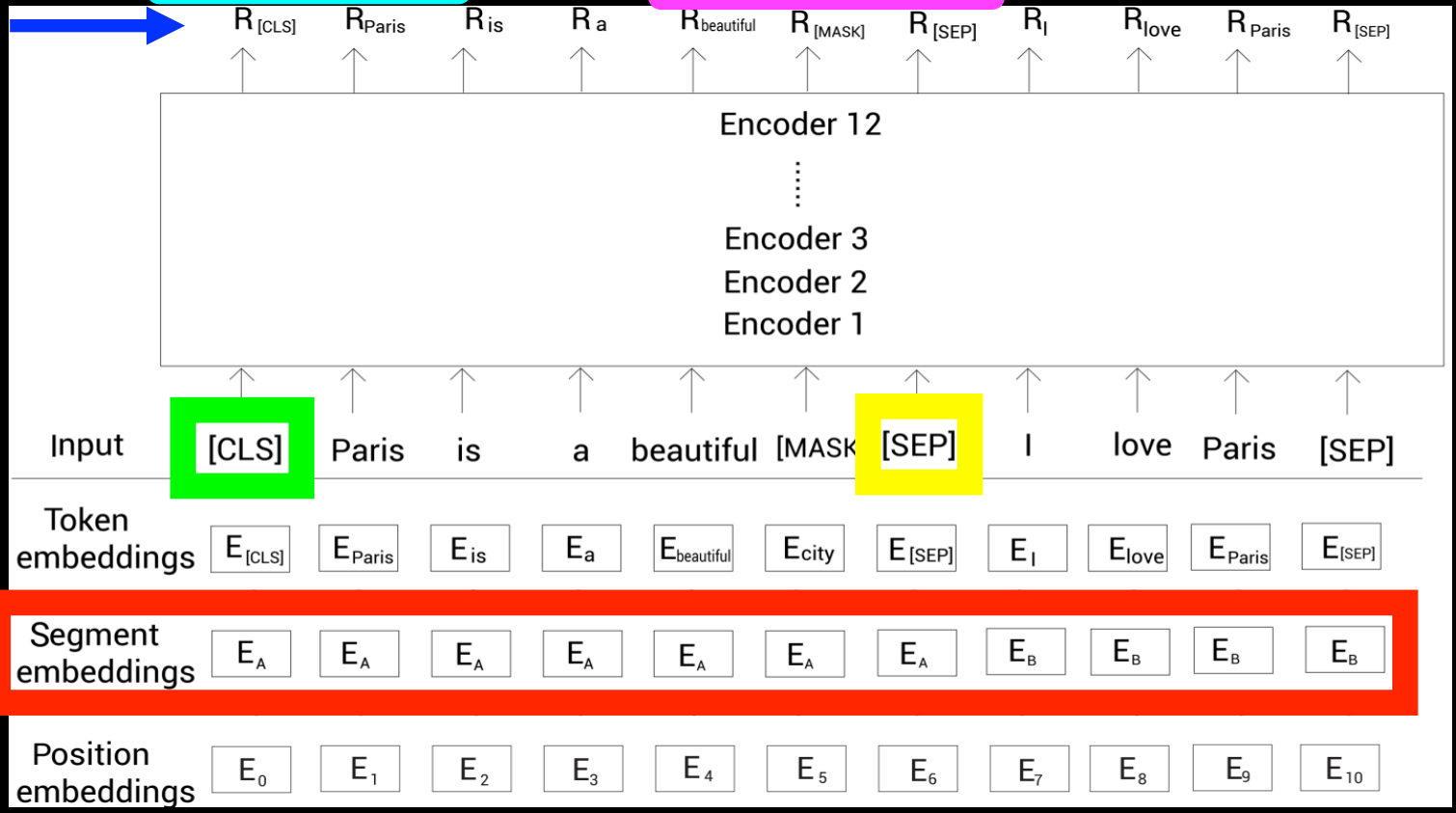
What is predicted?



What is predicted?



What is captured in the CLS token?



Two tokens introduced for 2nd task:

- CLS: representation of **ENTIRE** input
- SEP: separates two input sequences

Additional segment embedding signifies to which of two sentences a token belongs

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

Pretraining Details

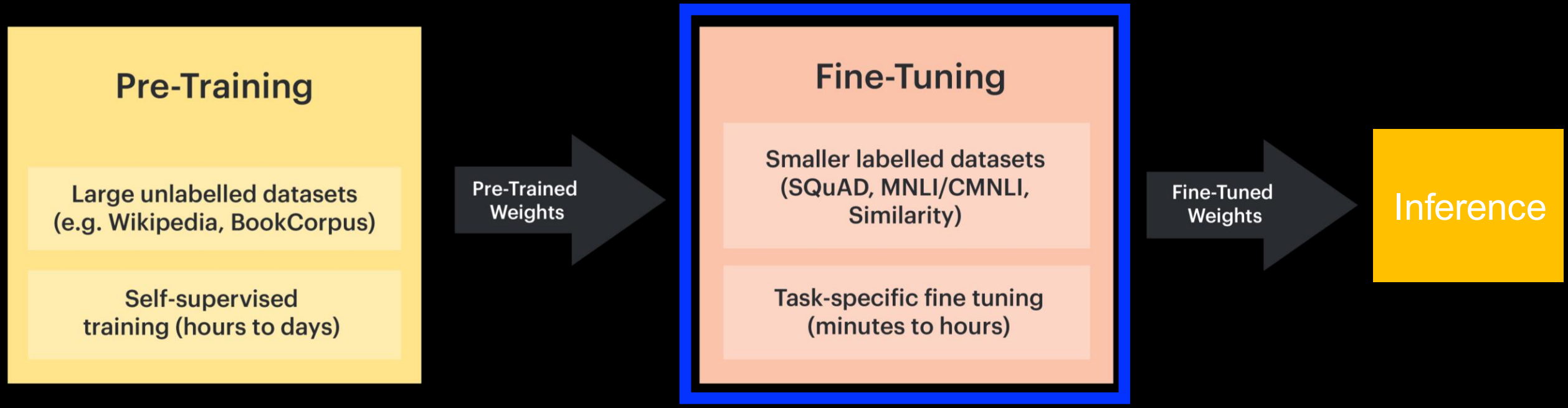
GPT

- Dataset: 800M words in BooksCorpus (>7,000 books)
- Weight initialization: Gaussian distribution for (0, 0.02)
- Optimizer: Adam
- Regularization: dropout and L2 norm penalty
- Batch size: 64 sequences of 512 tokens each

BERT Modifications

- Dataset: Extra 2.5B words in English Wikipedia
- Batch size: 256 sequences (two “sentences”) of 128 tokens

Approach

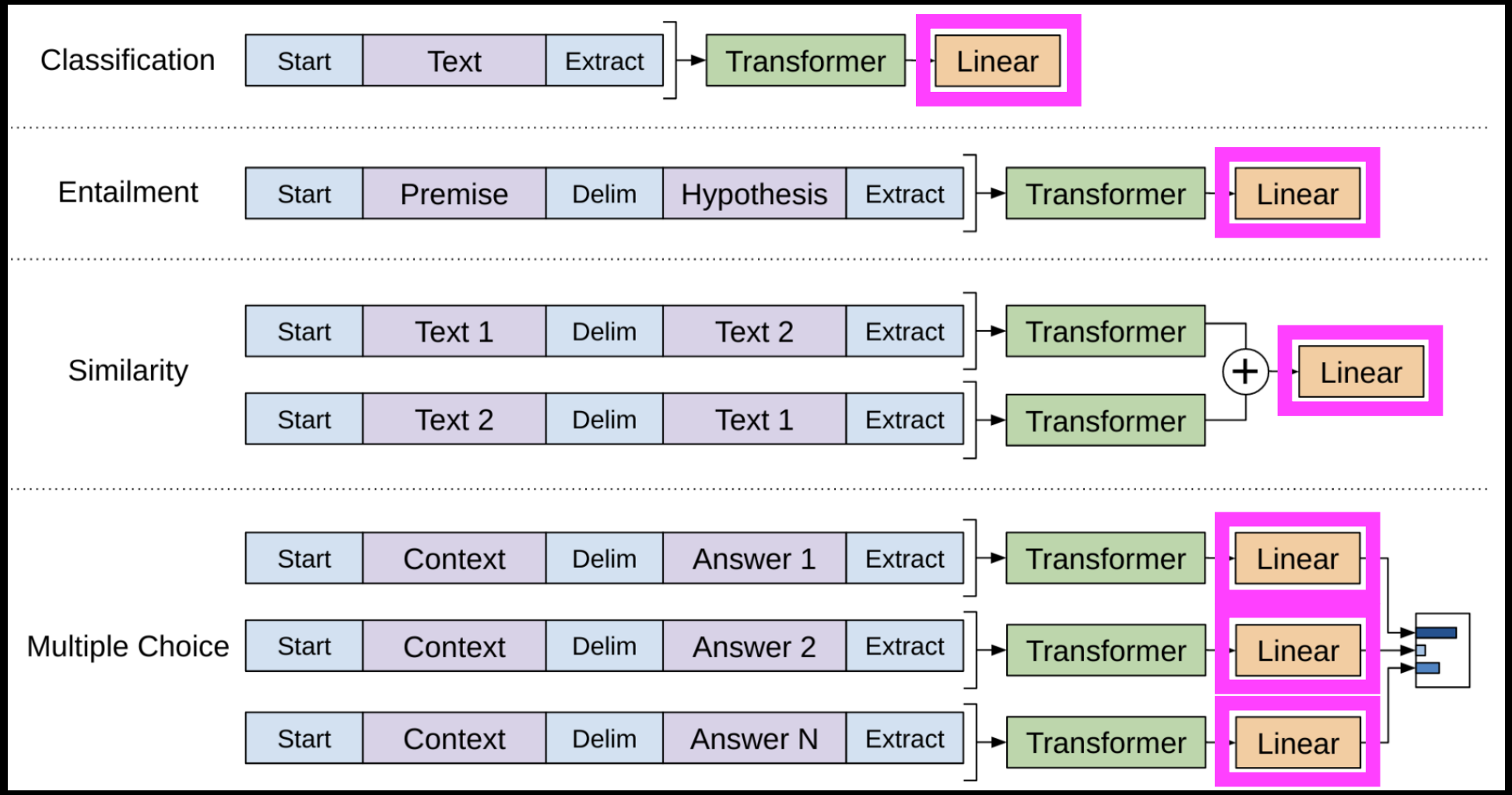


https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

GPT: Fine-Tuning to Four Types of Tasks (Discriminative Instead of Generative Tasks)

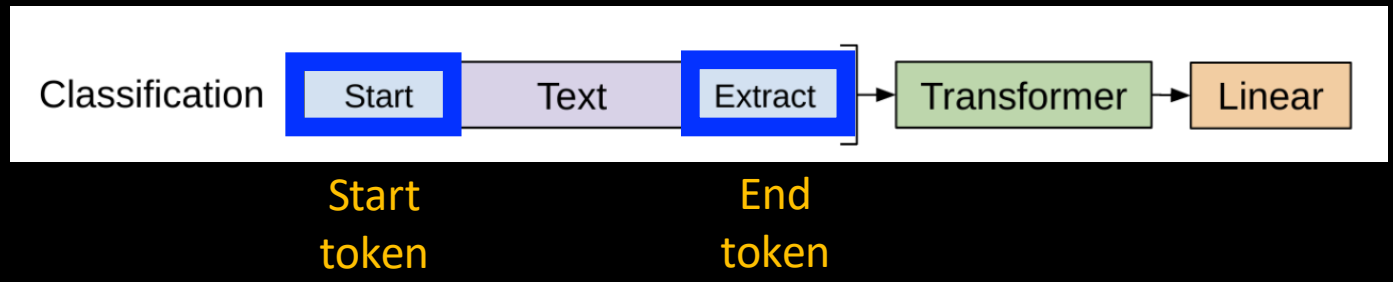
Append to pretrained transformer a small number of parameters (with softmax output), then perform supervised fine-tuning to task-specific datasets

Achieved state-of-the-art performance on 9 of 12 NLP dataset challenges!



Radford et al. Improving Language Understanding by Generative Pre-Training. Technical Report 2018.

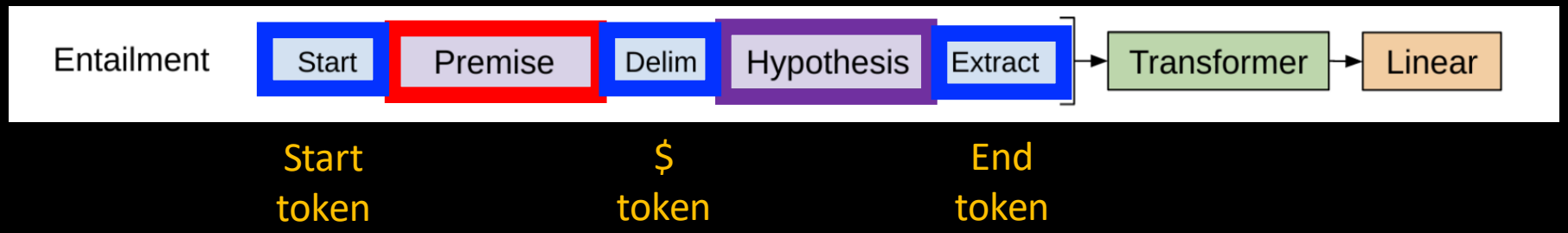
GPT: Fine-Tuning



e.g., predict whether a sentence from movie review has a positive vs negative sentiment

Radford et al. Improving Language Understanding by Generative Pre-Training. Technical Report 2018.

GPT: Fine-Tuning

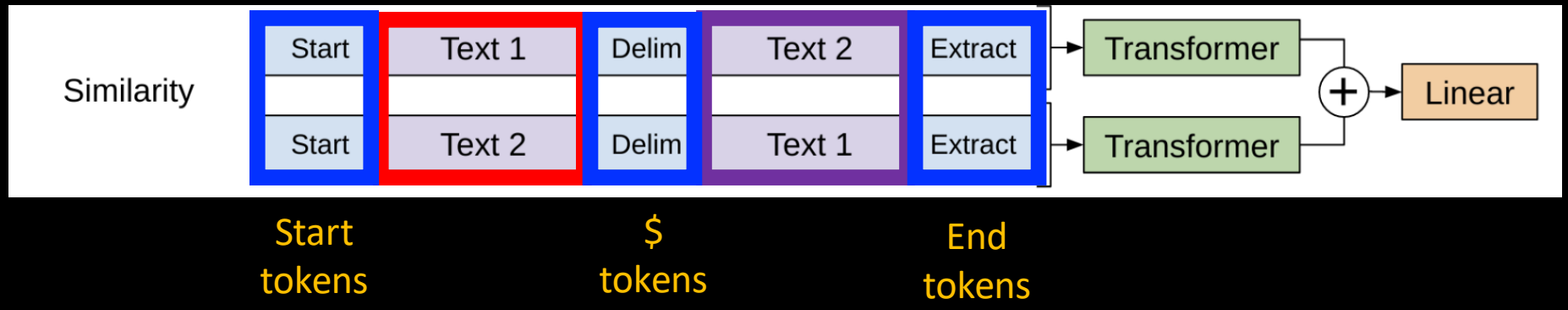


e.g., classify relationship of hypothesis to premise into three options;

Relationship	**Premise**	**Hypothesis**
Entailment	Students quietly listen to the lecture	The lecturer is speaking.
Neutral	Students quietly listen to the lecture	The projector equipment is broken.
Contradict	Students quietly listen to the lecture	The students are lecturing.

Radford et al. Improving Language Understanding by Generative Pre-Training. Technical Report 2018.

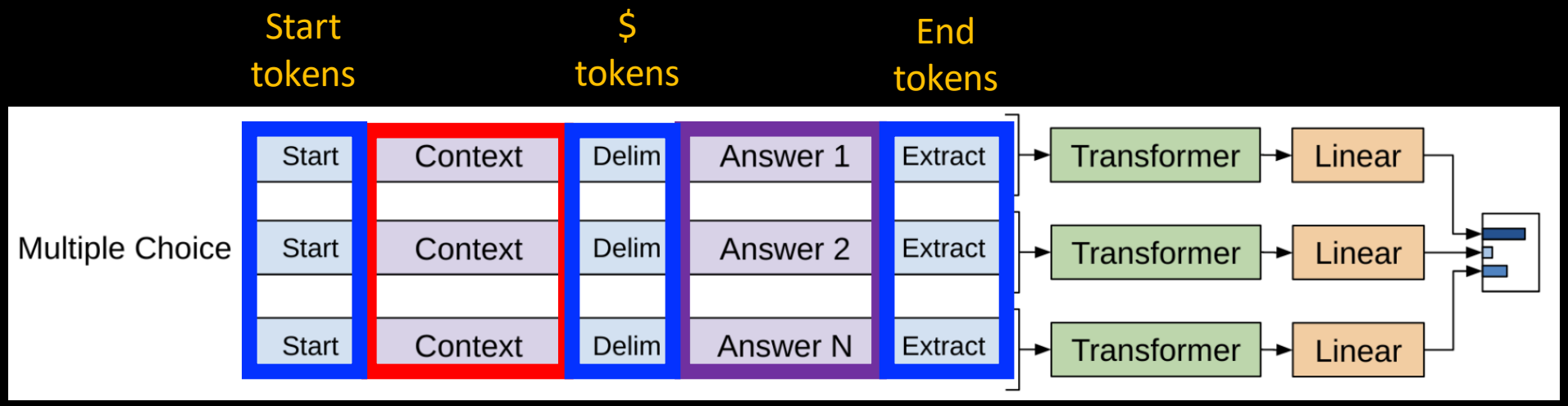
GPT: Fine-Tuning



e.g., decide if two sentences are similar (given no inherent ordering of the two sentences, both orders are used)

GPT: Fine-Tuning

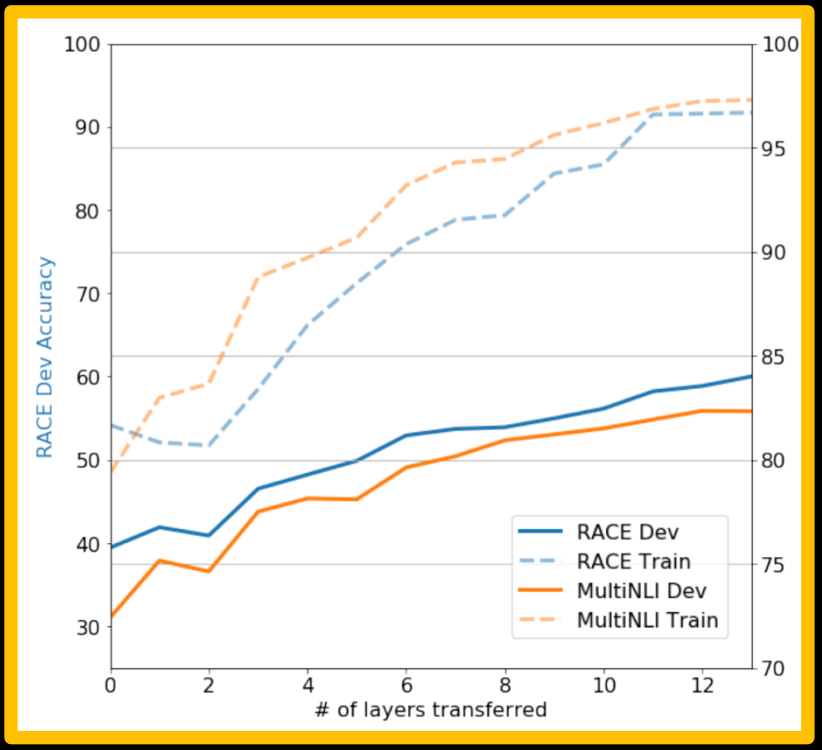
e.g., decide which answer results from the context (e.g., question about a document, concatenated); softmax layer gives probabilities for all options



Radford et al. Improving Language Understanding by Generative Pre-Training. Technical Report 2018.

GPT: Analysis of Design Choices

- How many pretrained layers should be fine-tuned?
- Does using a transformer rather than an LSTM help?
- Does pre-training help?

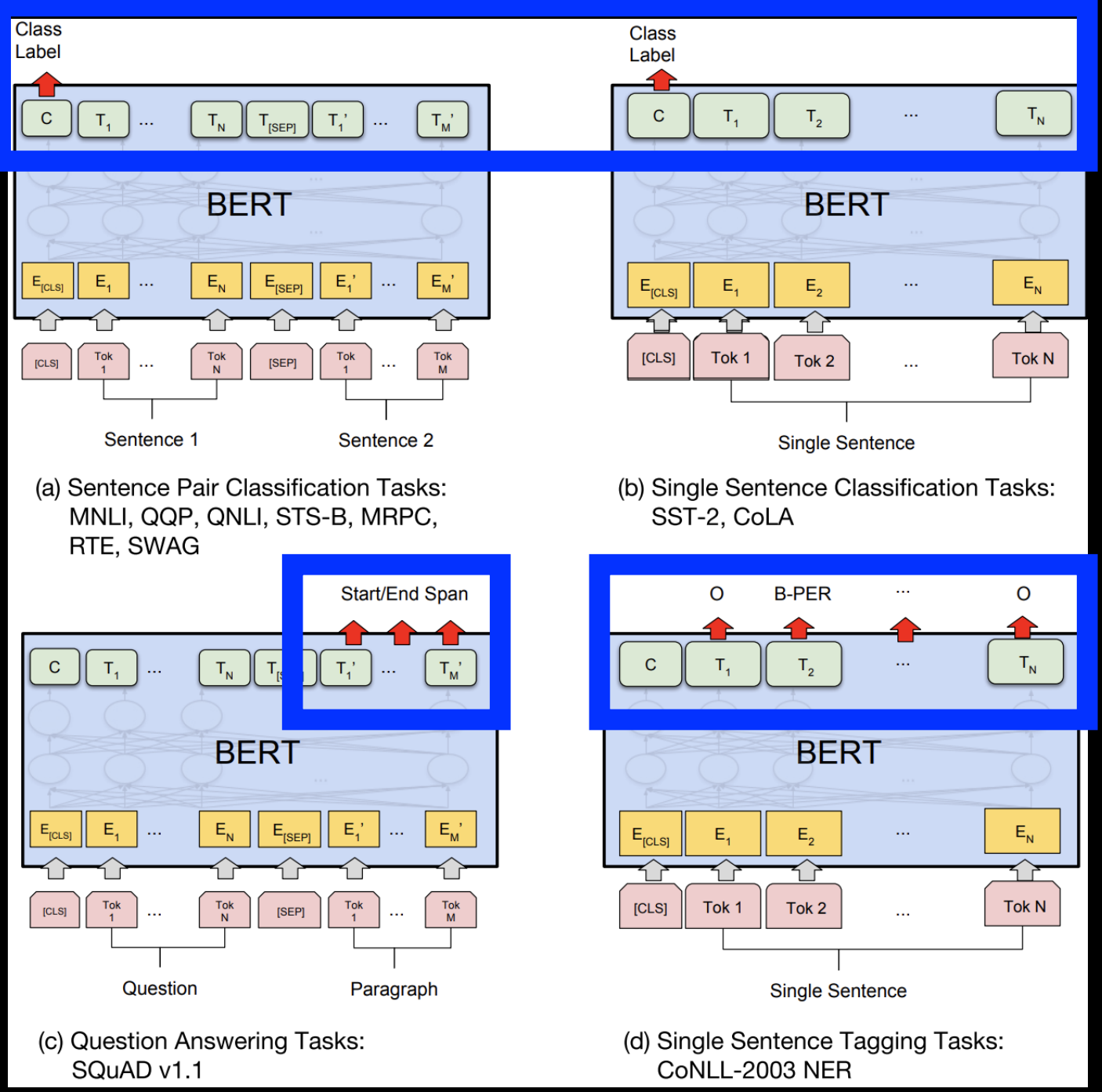


Method	Avg. Score	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	MNLI (acc)	QNLI (acc)	RTE (acc)
Transformer w/ aux LM (full)	74.7	45.4	91.3	82.3	82.0	**70.3**	**81.8**	**88.1**	**56.0**
Transformer w/o pre-training	59.9	18.9	84.0	79.4	30.9	65.5	75.7	71.2	53.8
Transformer w/o aux LM	**75.0**	**47.9**	**92.0**	**84.9**	**83.2**	69.8	81.1	86.9	54.4
LSTM w/ aux LM	69.1	30.3	90.5	83.2	71.8	68.1	73.7	81.1	54.6

Radford et al. Improving Language Understanding by Generative Pre-Training. Technical Report 2018.

BERT: Fine-Tuning to Three Types of Tasks

Achieved state-of-the-art performance on 11 NLP benchmark datasets!



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

(b) Single Sentence Classification Tasks: SST-2, CoLA

(c) Question Answering Tasks: SQuAD v1.1

(d) Single Sentence Tagging Tasks: CoNLL-2003 NER

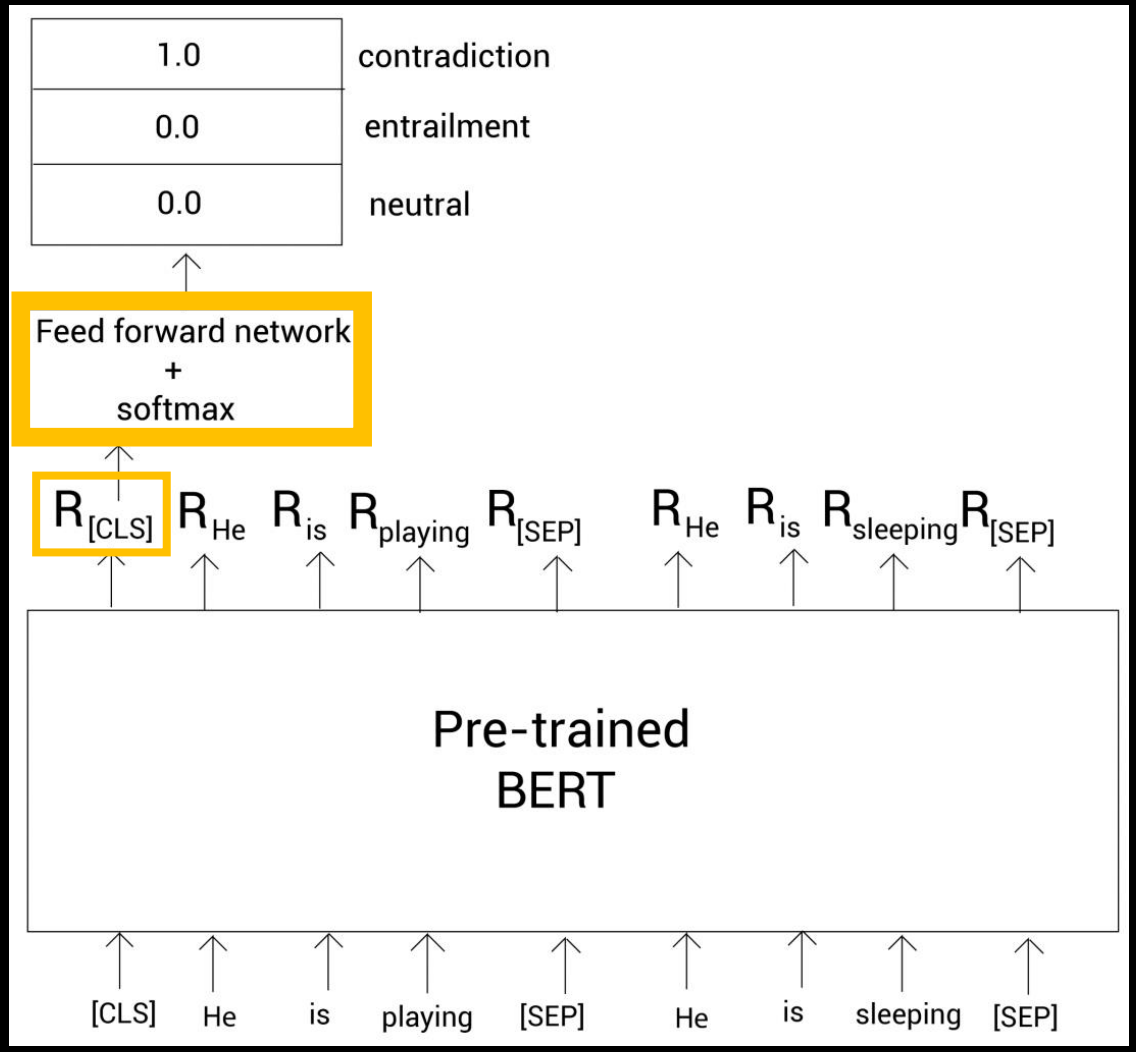
Devlin et al. BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding. ACL 2019.

BERT: Fine-Tuning for Classification

What is predicted?

Only new parameters

Predicts using the token representing ENTIRE input

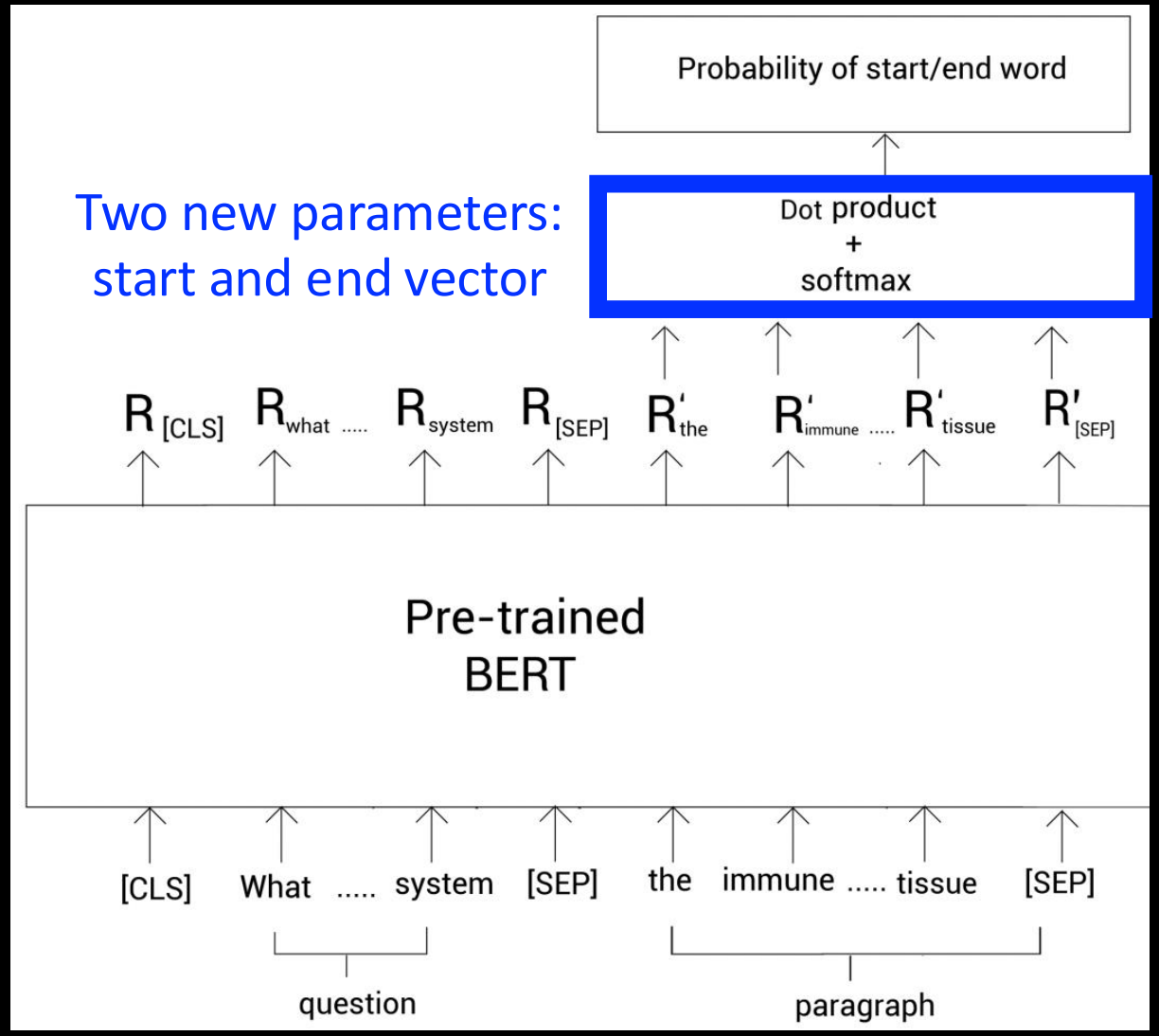


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BERT: Fine-Tuning for Question Answering

Predicts indices of start and end words in the input paragraph

- dot product comparing both learned vector representations with each input token

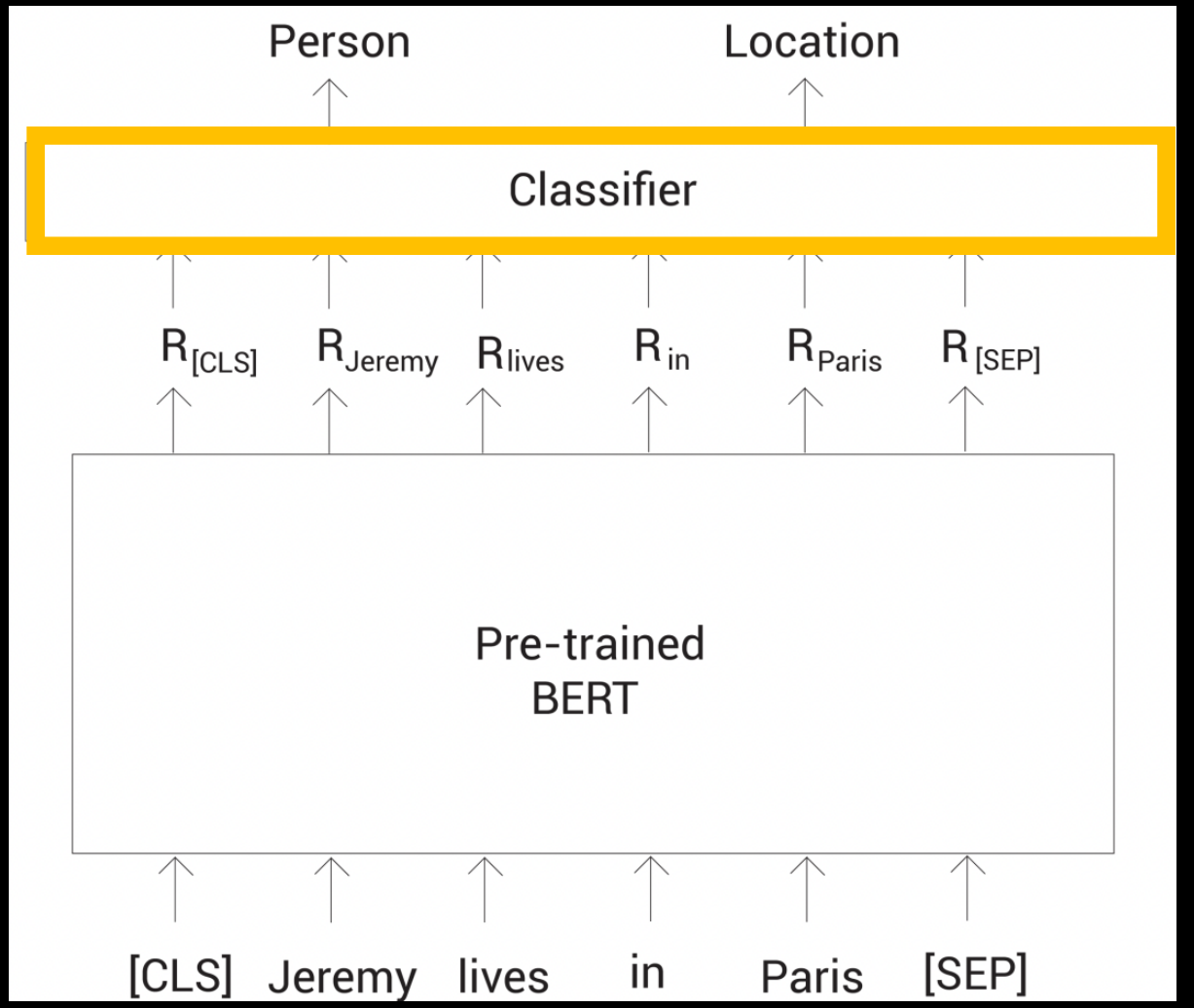


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BERT: Fine-Tuning for Single Sentence Tagging

Only new parameters

Each token's new representation is passed to classifier (e.g., named entity recognition)



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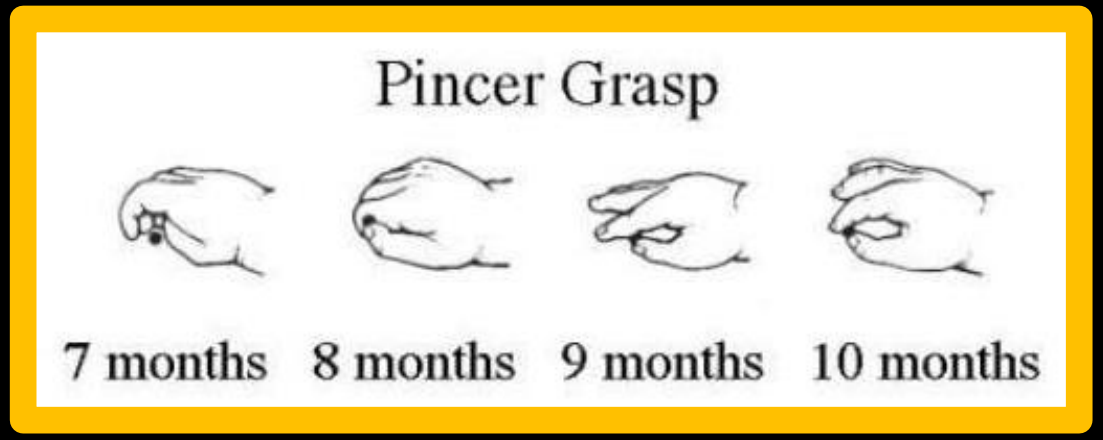
BERT: Analysis of Design Choices

- Does including next sentence prediction in pretraining help?
  - Yes
- Does including masked token prediction in pretraining help? *(versus using GPT's left-to-right approach instead)*
  - Yes; worse results from unidirectional than bidirectional pretraining

Tasks	Dev Set MNLI-m (Acc)	QNLI (Acc)	MRPC (Acc)	SST-2 (Acc)	SQuAD (F1)
$\text{BERT}_{\text{BASE}}$	84.4	88.4	86.7	92.7	88.5
No NSP	83.9	84.9	86.5	92.6	87.9
LTR & No NSP	82.1	84.3	77.5	92.1	77.8

Devlin et al. BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019.

Summary: General-Purpose Representation Pre-Training Outperforms Task-Focused Learning



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(EMNLP, 2019)

LXMERT: Learning Cross-Modality Encoder Representations from Transformers

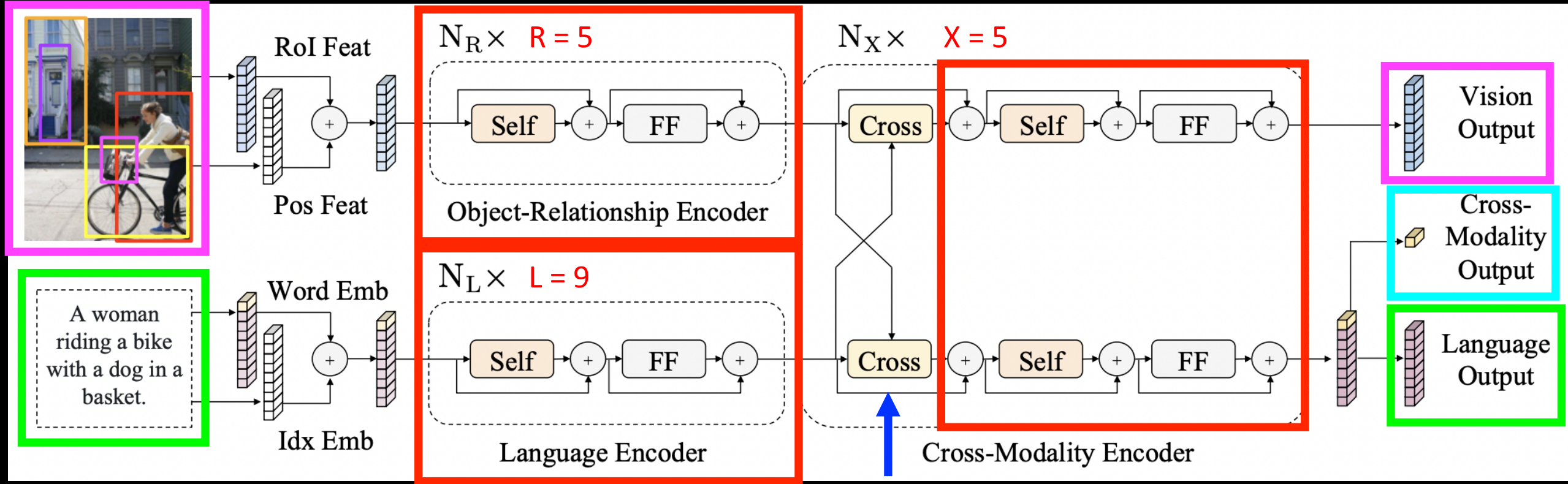
Hao Tan **Mohit Bansal**
UNC Chapel Hill

Architecture: Embeds BERT

What is captured in representations for:
- Vision?
- Language?
- Cross-modality ([CLS] token)?

Faster R-CNN features for *36* objects detected from 1,600 categories

BERT (Encoder)



WordPiece subword tokenization plus [CLS]

What do the new representations capture for: Vision? Language?

What is captured in the new representations from cross-attention (i.e., "query" from one modality with "keys" and "values" from the other)?

Tan and Bansal, EMNLP 2019.

Pretraining: Data (Larger Scale)

Image Split	Images	Sentences (or Questions)					
		COCO-Cap	VG-Cap	VQA	GQA	VG-QA	All
MS COCO - VG	72K	361K	-	387K	-	-	0.75M
MS COCO ∩ VG	51K	256K	2.54M	271K	515K	724K	4.30M
VG - MS COCO	57K	-	2.85M	-	556K	718K	4.13M

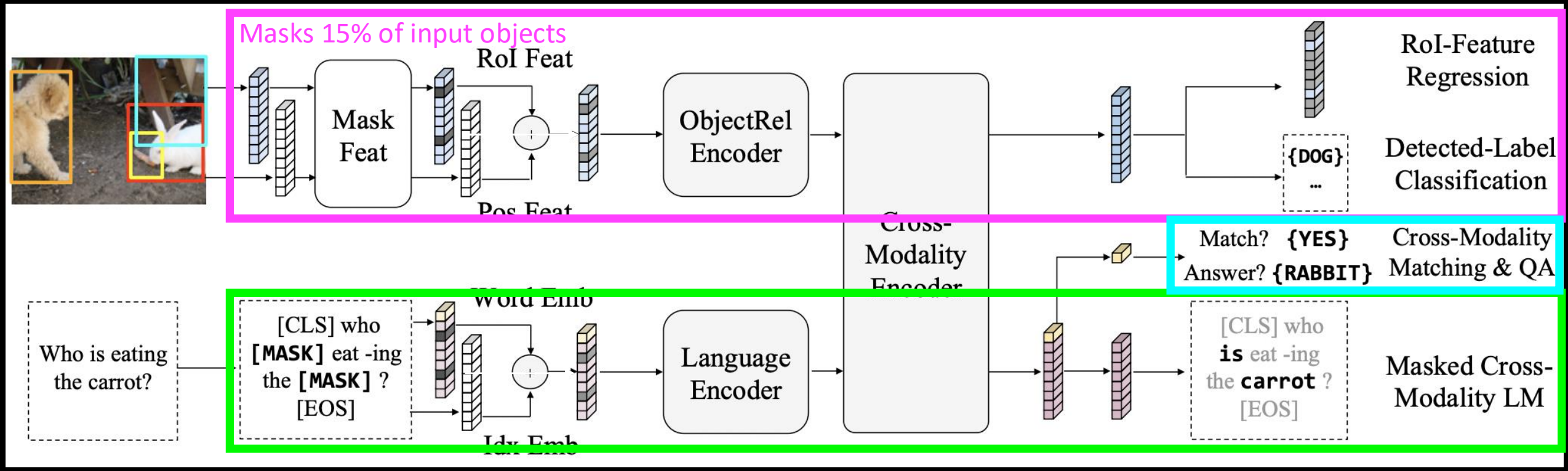
9.18M image-sentence pairs for 180,000 images (questions in VQA datasets used for image-sentence pairs) coming from 2 image captioning and 3 VQA datasets

Tan and Bansal, EMNLP 2019.

Pretraining: Objective (5 Losses Summed)

(Faster R-CNN detector frozen during training)

* 50% of language responses are chosen at random



Training took 10 days on 4 Titan Xp GPUs

Tan and Bansal, EMNLP 2019.

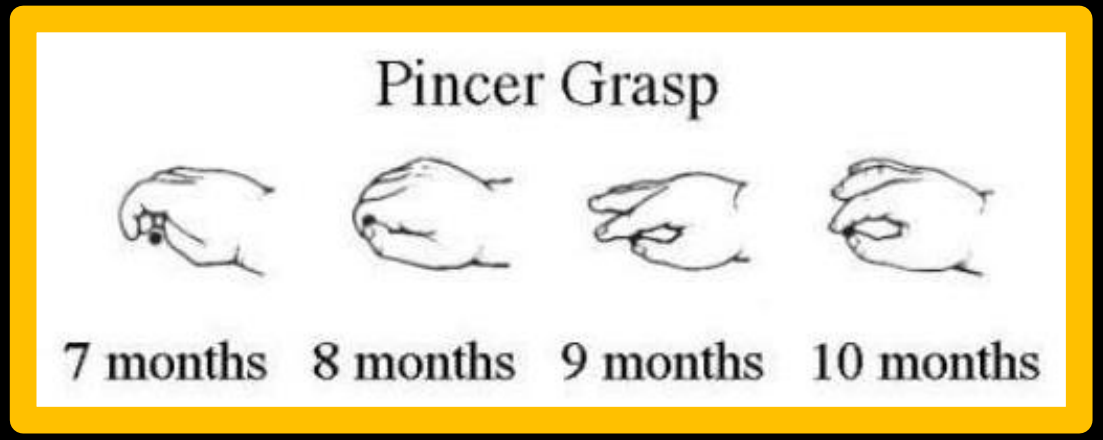
Fine-Tuning: Experimental Results

Method	VQA				GQA			$NLVR^2$	
	Binary	Number	Other	**Accu**	Binary	Open	**Accu**	Cons	**Accu**
Human	-	-	-	-	91.2	87.4	89.3	-	96.3
Image Only	-	-	-	-	36.1	1.74	17.8	7.40	51.9
Language Only	66.8	31.8	27.6	44.3	61.9	22.7	41.1	4.20	51.1
State-of-the-Art	85.8	53.7	60.7	70.4	76.0	40.4	57.1	12.0	53.5
LXMERT	**88.2**	**54.2**	**63.1**	**72.5**	**77.8**	**45.0**	**60.3**	**42.1**	**76.2**

State-of-the-art performance for 2 VQA datasets and 1 visual reasoning dataset (i.e., does statement describe two images or not?)

Tan and Bansal, EMNLP 2019.

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(NeurIPS, 2020)

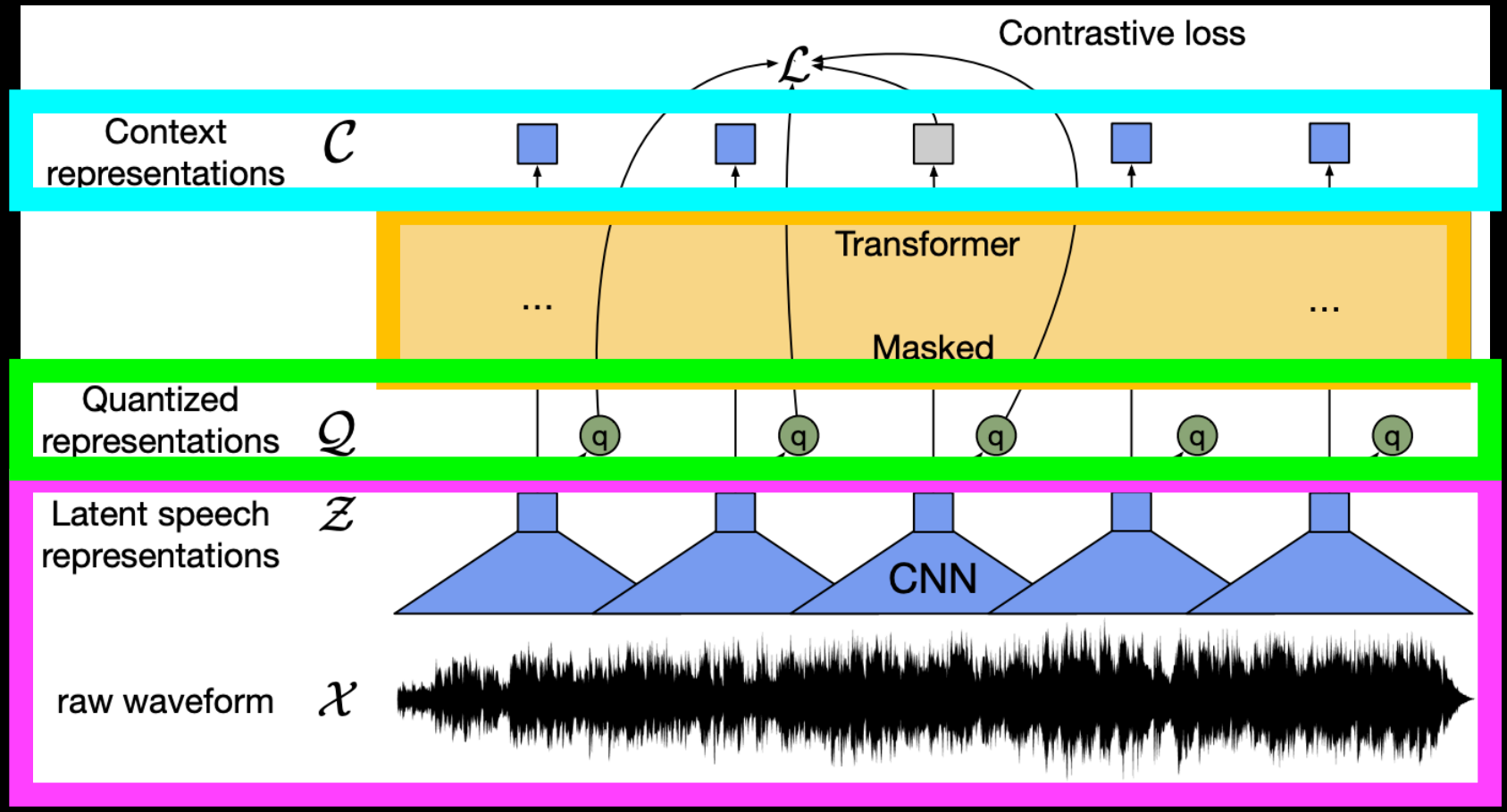
wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations

Alexei Baevski **Henry Zhou** **Abdelrahman Mohamed** **Michael Auli**

{abaevski,henryzhou7,abdo,michaelauli}@fb.com

Facebook AI

Architecture: Embeds BERT



Baevski et al. NeurIPS 2002.

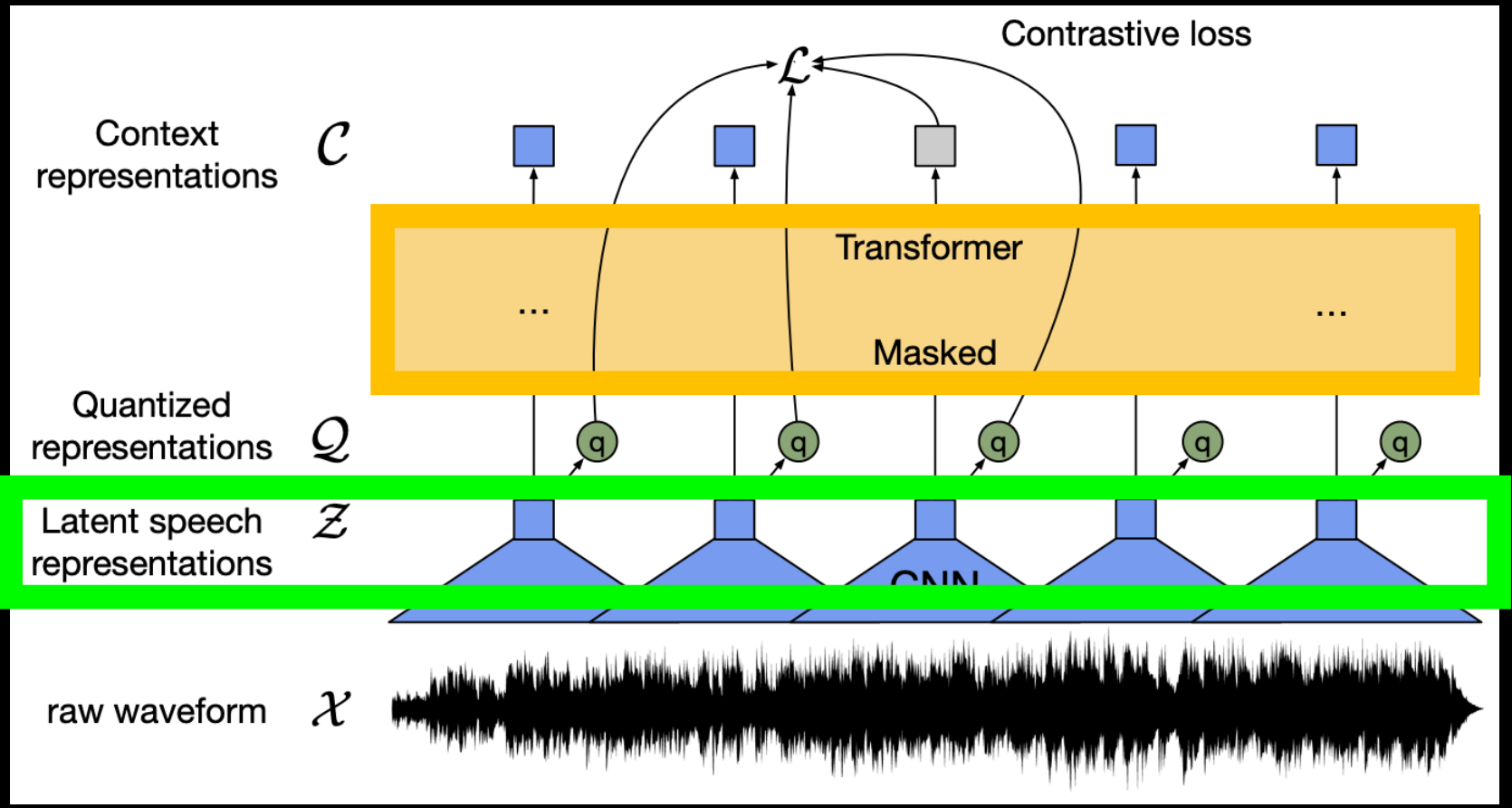
What do the new representations capture?

Representations discretized to finite set

Input representation for transformer learned from raw audio; captures short durations for input

What happens to the number of inputs if the CNN's stride increases?

Pre-Training: Masked Token Prediction (from a Set of Candidates)



Masks these tokens

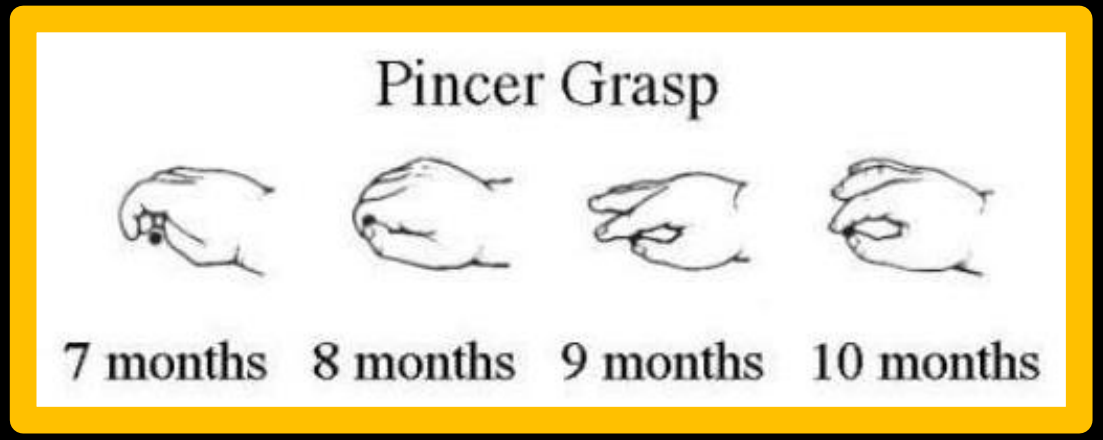
Training data: 2 datasets of 960 hours and 53.2k hours of audio respectively

Baevski et al. NeurIPS 2002.

Fine-Tuning: Experimental Design and Results

- Adds a linear layer for predicting target categories (e.g., 30 tokens for Librispeech)
- Achieved state-of-the-art performance for speech recognition (Librispeech benchmark) and phoneme recognition (TIMIT benchmark), including when using 100 times less labeled data
- Achieved word error rate of 4.8 when using only 10 minutes of labeled data
  - Relevant for many of the ~7,000 languages spoken worldwide, which often lack thousands of hours of transcribed speech

Summary: General-Purpose Representation Pre-Training Outperforms Task-Focused Learning



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/



Today's Topics

- Revisiting learning: from solving tasks to building representations that generalize
- The recipe: large-scale pretraining + lightweight fine-tuning
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- Speech test: can the recipe replace engineered solutions for speech recognition?
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- Programming tutorial

(ICLR, 2021)

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]equal technical contribution, [†]equal advising

Google Research, Brain Team

Architecture: BERT

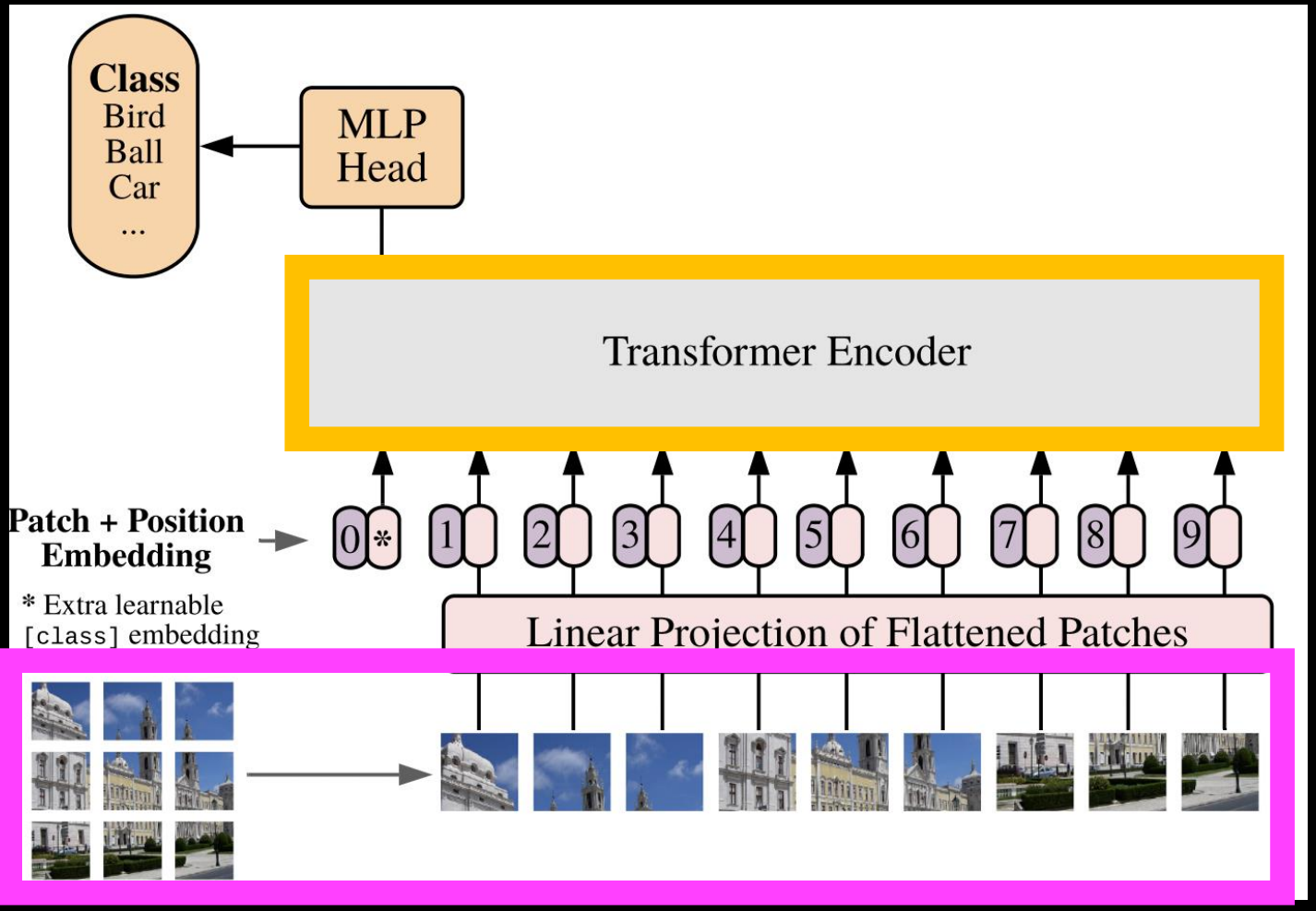


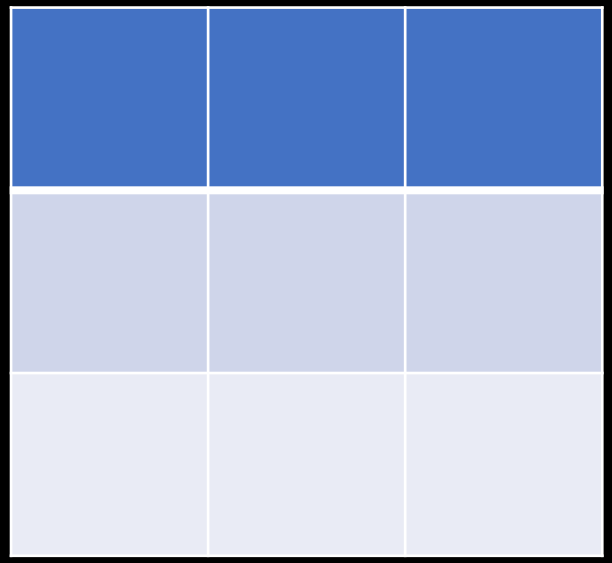
Image decomposed into 16x16 patches (example simplified); representations include “flattened” and ResNet features

Why not use the raw pixels as input?

Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR 2021.

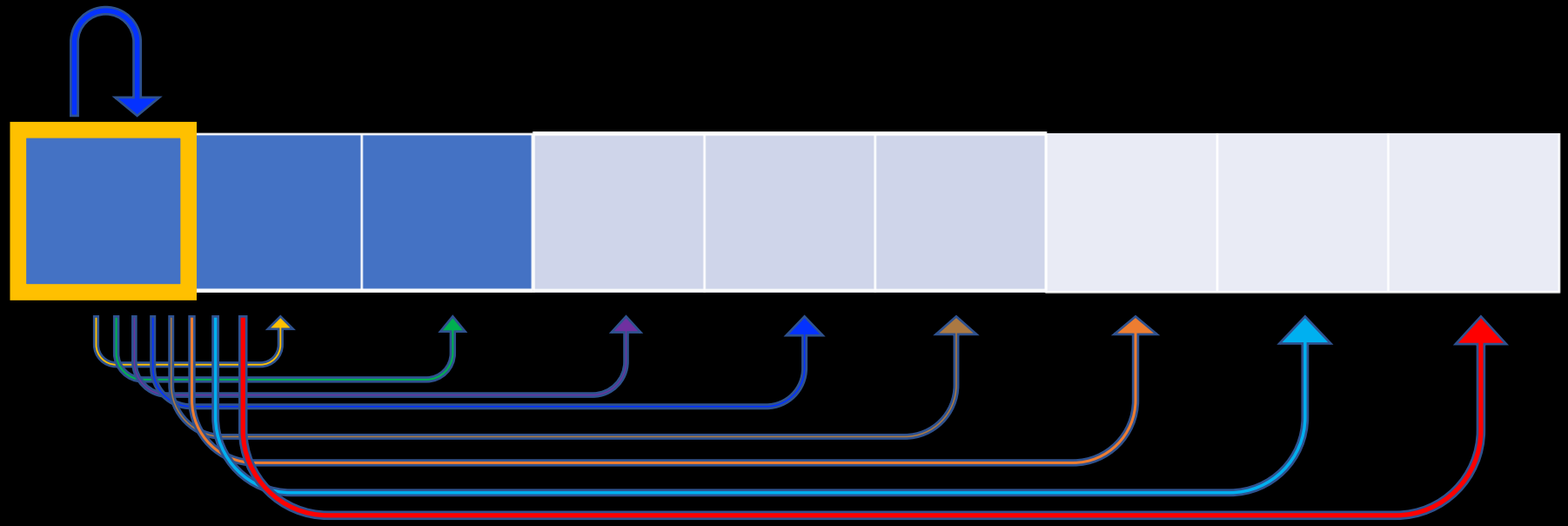
Recall Self-Attention: Idea

New representation of each pixel showing its relationship to all pixels; e.g., assume this 3x3 image



Recall Self-Attention: Idea

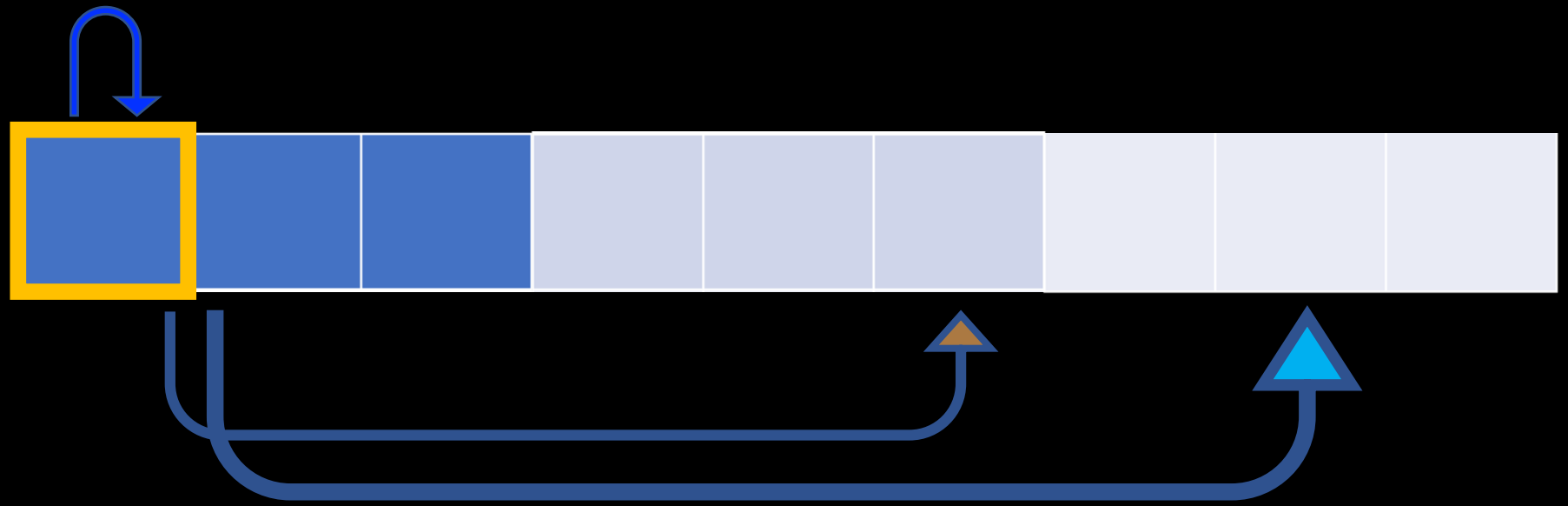
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Learned new representation indicates which global information clarifies a pixel's meaning (e.g., include in the representation of a pixel of an eye context of what animal it belongs to)

Recall Self-Attention: Idea

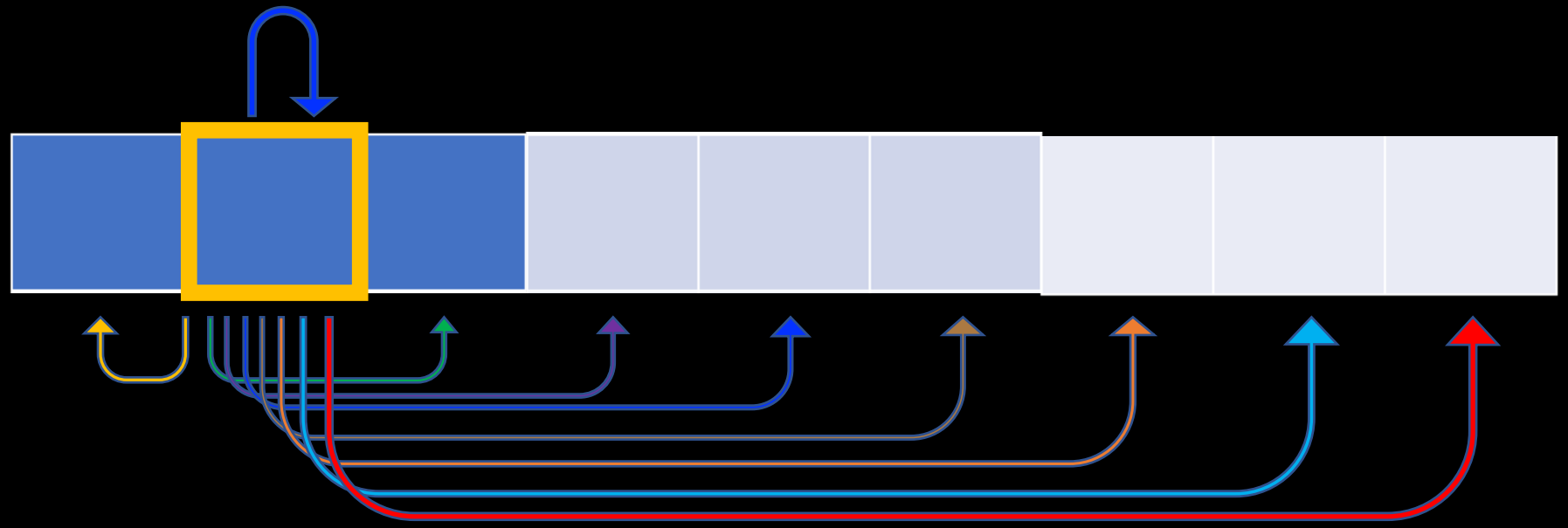
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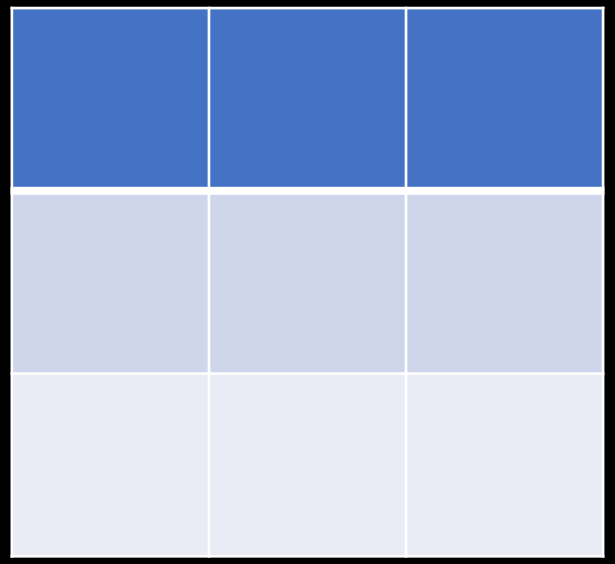


And so on for remaining image pixels…

Rationale for Patches: Computational Cost of Self-Attention

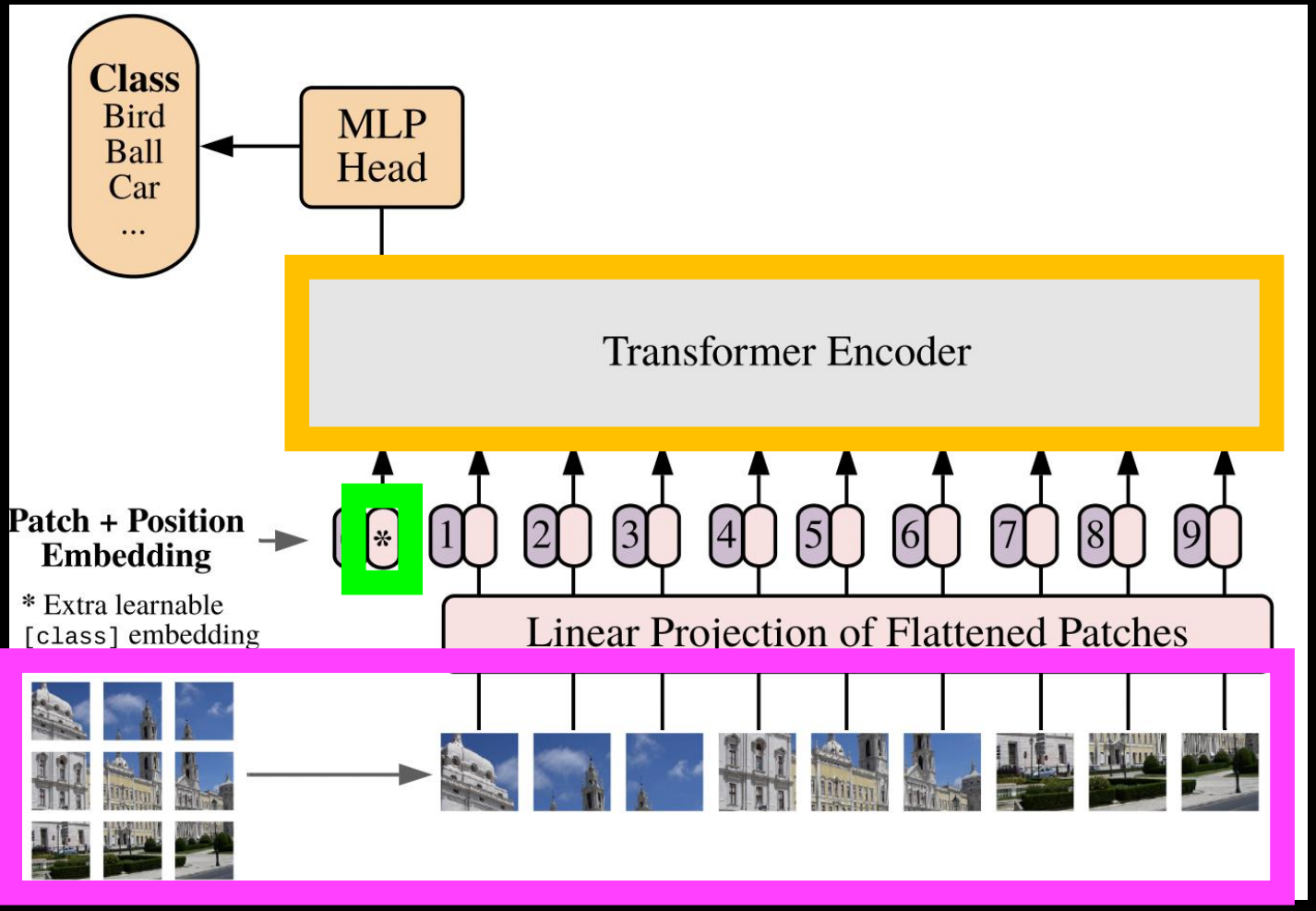
e.g., instead of using 3x3 image, what if a 1920 x 1080 image was used? How many self-attention computations would be needed?

- $(1920 \times 1080)^2 = 4{,}299{,}816{,}960{,}000$ (i.e., ~4.3 trillion)



Quadratic cost of self-attention is often impractical for pixels!

Architecture: BERT



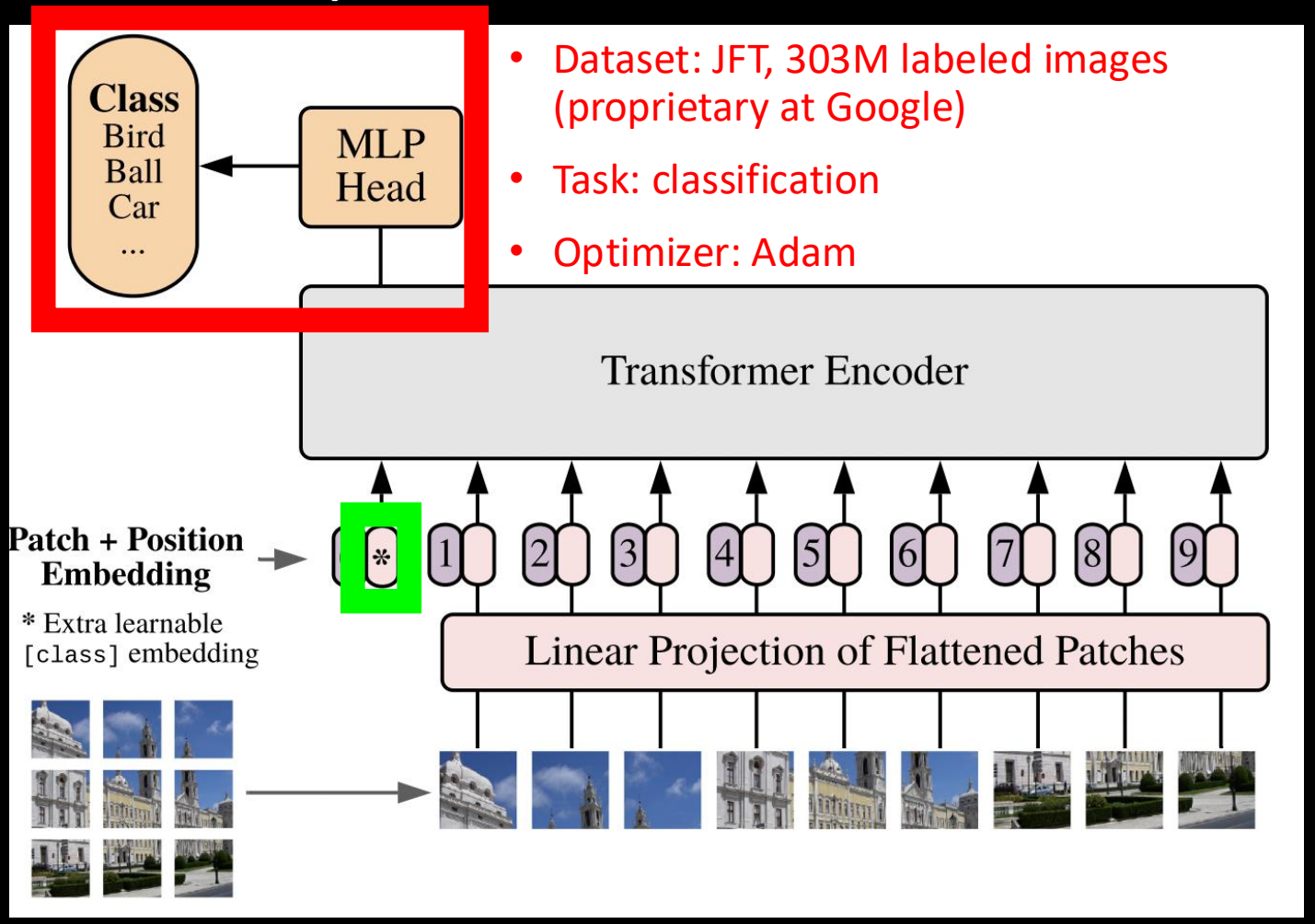
What does the CLS token represent, after passing through the transformer?

Image decomposed into 16x16 patches

For a 160 x 160 pixel image, how many patches (inputs) would be extracted?
- 100

Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR 2021.

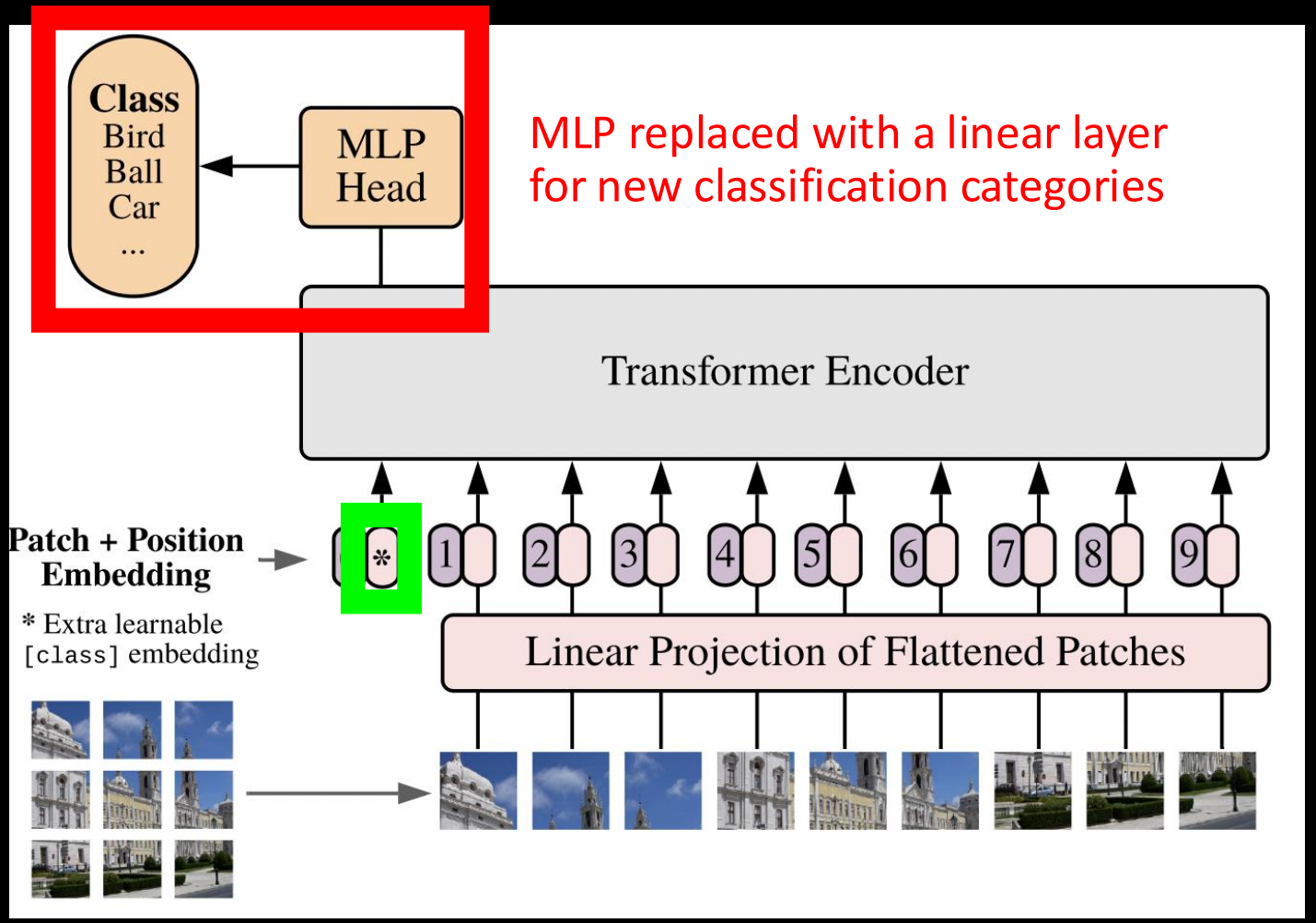
Pre-Training: Supervised Learning Beats Self-Supervised Masked Prediction



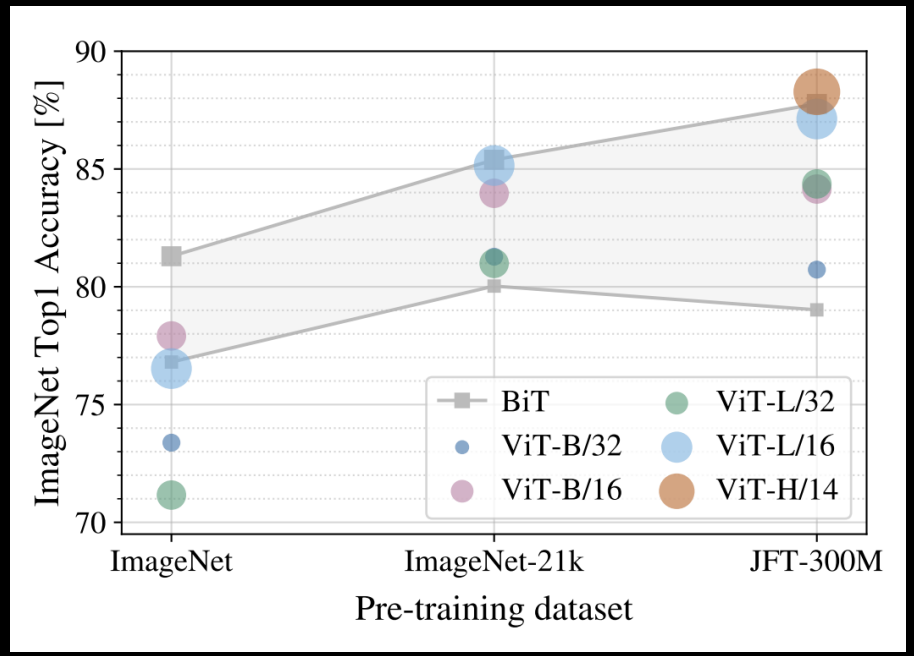
- Dataset: JFT, 303M labeled images (proprietary at Google)
- Task: classification
- Optimizer: Adam

Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR 2021.

Fine-Tuning: Other Image Classification Tasks



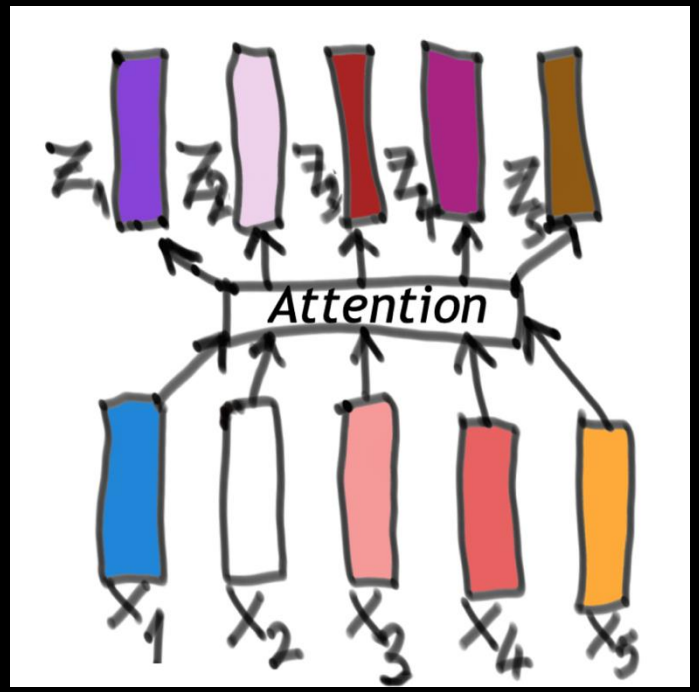
Achieved strong results on five image classification datasets!



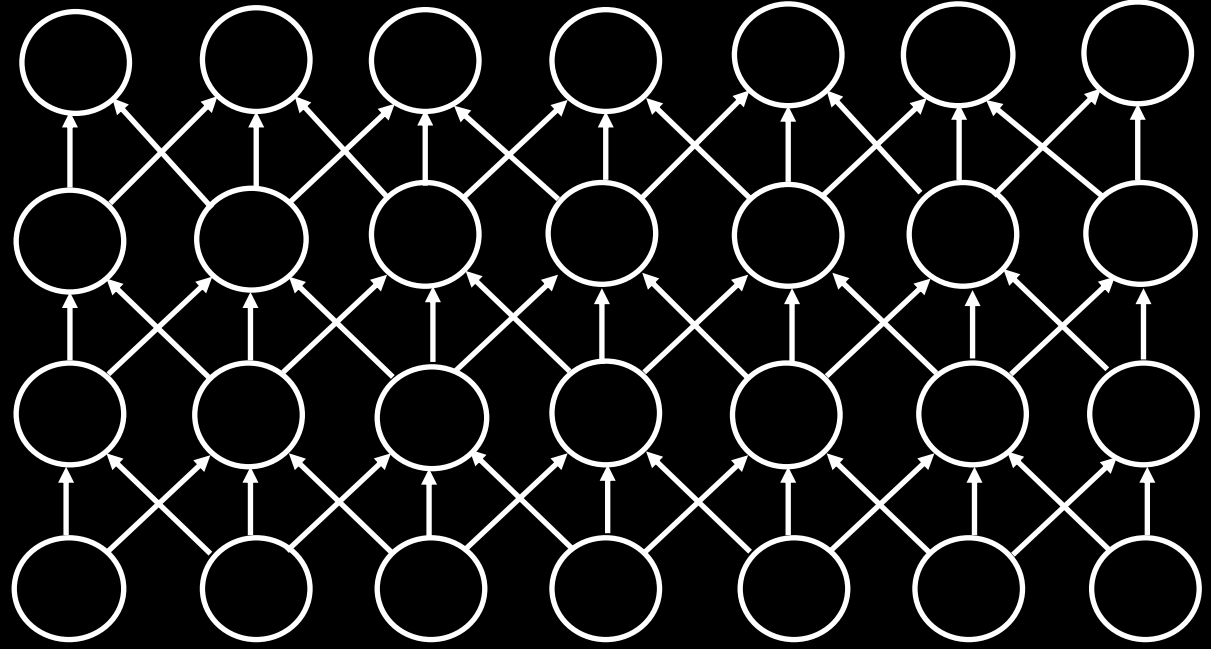
Pretraining dataset size was critical, regardless of model size (ImageNet fine-tuning results)

Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR 2021.

Finding: Large-Scale Pretraining Enables Transformers to Compete with CNNs



https://towardsdatascience.com/self-attention-5b95ea164f61

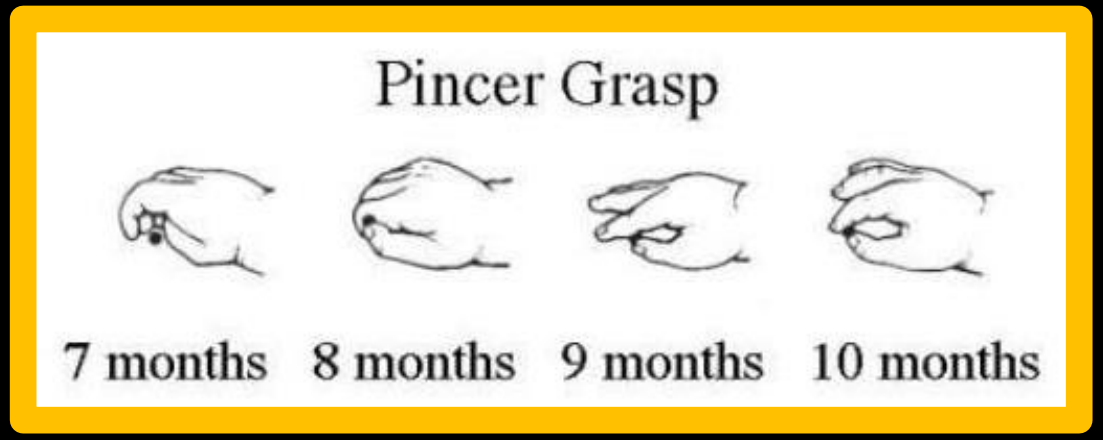


https://www.deeplearningbook.org/contents/convnets.html

Self-attention: each layer has a global receptive field

Convolutional layers: deeper layers have increasingly more global receptive fields

Summary: General-Purpose Representation Pre-Training Outperforms Task-Focused Learning



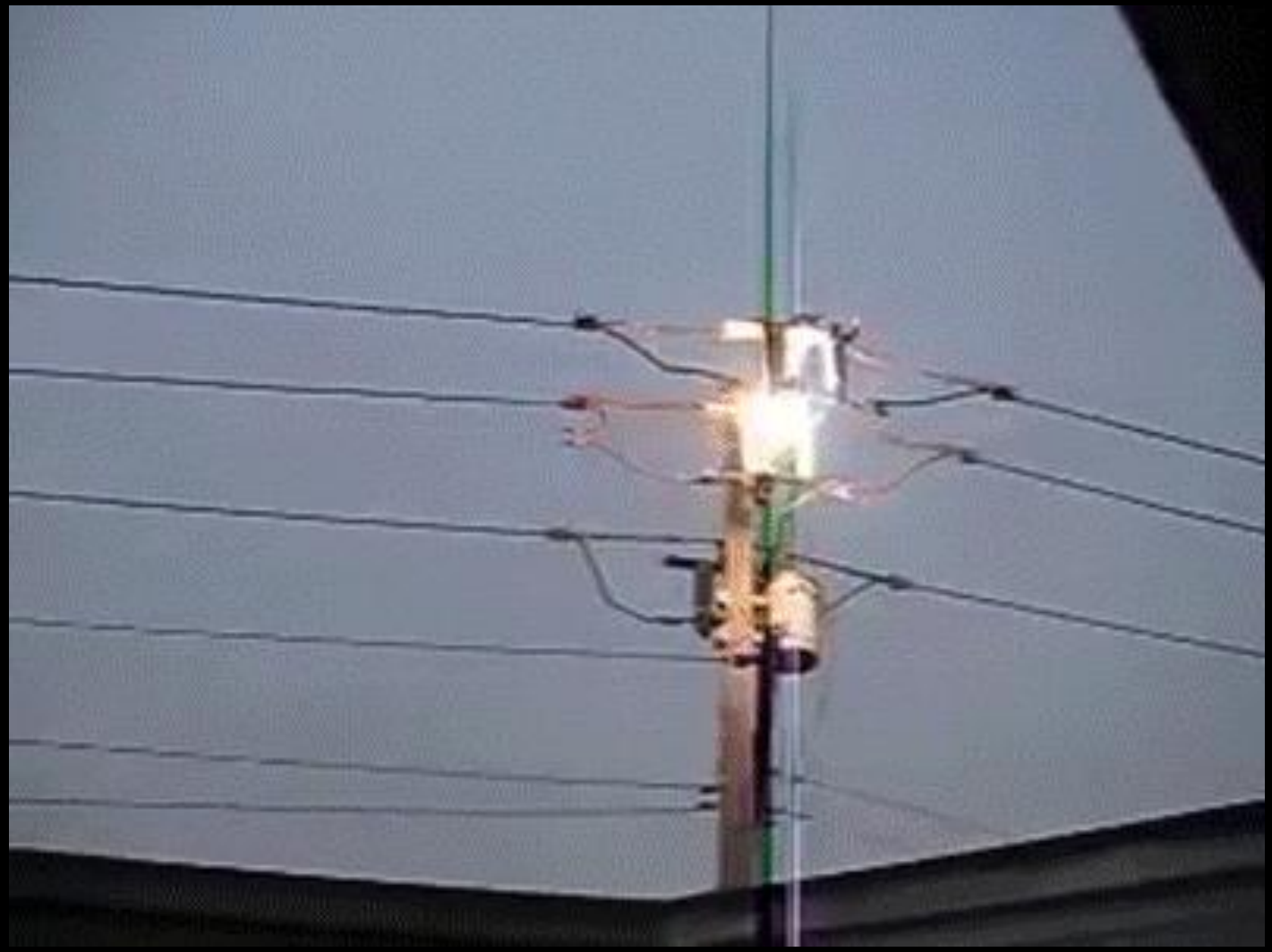
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What’s Next: Explosion of Transformers in Society



https://www.pinterest.com/pin/521784306804400819/

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The End